

Supplementary Information

Wheat phenology partly offsets projected increases in Septoria damage across Europe

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Supplementary Methods

Crop, disease and yield relationships

Daily climate data drive wheat-phenology and Septoria-disease calculations. Wheat phenology determines leaf appearance, unfolding and the flowering-to-soft-dough interval. The disease model predicts damage on those leaves; its damage does not feed back into crop-development rates. After symptom detection, damage is mapped to an assumed loss of leaf function. Integration gives HAD loss, and a published coefficient converts HAD loss to a conditional disease-related yield estimate. This final calculation is separate from a complete crop-yield model.

The reference upper-three-leaf canopy has maximum LAI of one. Disease-induced HAD loss integrates the assumed loss of functional green leaf area over time, with units of GLAI-days. Division by maximum reference LAI gives normalized HAD loss, expressed as equivalent days of reference canopy function lost. Numerical values are identical in the unit-LAI scenario. Relative HAD loss is the ratio of HAD loss to reference HAD and describes loss of assumed canopy function over the integration interval. Measured leaf-lesion percentage has a different denominator and temporal meaning.

Full-grid climate estimates

The experiment includes all 14,941 wheat land-use cells, with eligible calendars for 14,932 cells. Invalid seasons have unavailable outcomes. Symptom timing requires detection, while symptom frequency includes complete seasons without symptoms. Each metric has its own valid historical-future pairs. Corresponding positions in the two 30-year periods are paired within a climate model and SSP.

Spatial means divide the harvested-area-weighted sum of paired changes by the harvested-area-weighted number of valid pairs. Historical and future means use these same pairs. Climate-model ranges span three independently aggregated estimates. The full-grid calculation has no spatial sampling error. Environmental regions follow EEA centroid assignment and countries follow dominant SPAM source-country labels; outside-region and unassigned cells remain explicit.

Field observations and prediction scores

Pooled disease evaluation excludes calibration records and combines geographically eligible records from both archives. Source identifiers and assessment definitions remain in the evidence files. Calibration and evaluation have no shared field identifiers. Coordinate-year identities define geographic bootstrap groups, with overlapping identities treated as the same group.

An assessment weight is the reciprocal of the number of coordinate-years multiplied by the reciprocals of fields, source leaves and observations within successive groups. RMSE is the square root of the weighted mean squared error. Twenty thousand paired coordinate-year bootstrap resamples, with seed 20261008, give 95% percentile intervals. Every resample retains both predictions and all field, leaf and date records in a sampled group. The intervals exclude parameter-selection uncertainty and verification of independent biological-site identities.

Symptom-onset comparisons use observed intervals bracketing first symptoms and specified penalties for missed positive observations. Crop-stage evaluation also uses two-sided observation intervals; plotted interval centres are display locations, not exact observed event

dates. Calibration profiles and nested model selection are separate from pooled evaluation. German station observations evaluate the established phenology formulation; field support for added upper-leaf stages and soft dough remains limited.

Canopy-based yield estimates

Published protected and unprotected treatment means retain their leaf coverage, growth-stage interval, cultivar and disease exposure. The eight contrasts across five backgrounds include two with STB alone, two with yellow rust and four with mixed diseases. Absolute and relative yield scores describe aggregate treatment means when each genetic background is excluded from fitting. Raw plot variation and covariance are unavailable.

The grid yield estimate is a linear conversion of HAD loss under the assumed upper-three-leaf canopy. Fixed coefficients and alternative GS31–85 and GS65–85 intervals assess dependence on these assumptions. The primary interval is shorter than the source coefficient interval, so the conversion requires field validation. Positive projected yield changes denote a decrease in the estimated disease-loss component; they do not establish a gain in total yield under future climate.

Severity-based field yield comparisons

The consolidated Tunisia/Nordic inventory contains 423 dated leaf or treatment assessments sharing 272 yield keys. Exclusion of a known control alias and post-harvest assessments leaves 412 assessments and 263 outcomes. These inventory counts do not denote independent environments. The Nordic predictive population contains 62 treated–untreated contrasts in five trials; subsets with known assessment stage and an earlier matched assessment contain 51 and 40 contrasts, respectively. Missing treated-plot stem-extension assessments remain unavailable. Growth-stage interactions use recorded ordinal GS, rather than elapsed time. The fixed stage-regression basis is severity reduction multiplied by $(GS - 75)/10$. The secondary nonnegative basis uses severity reduction multiplied by $(85 - GS)/10$. Neither basis represents observed infection timing.

The Tunisian plot table contains incidence-weighted lesion area on flag and immediately lower leaves. Flag leaves were assessed at GS75 in both years; lower leaves were assessed at GS61 in 2018 and GS73 in 2019. The leaf-stage and year differences prevent separate identification of leaf-specific physiological coefficients. No same-leaf AUDPC or HAD is reconstructed. Cultivar proportions define composition predictors and shared mixture identities; treatment numbers are year-specific. Two-way whole-year transfer evaluates absolute yield on 82 outcomes. The 40 source-matched reference contrasts retain one negative response; within-year validation excludes entire mixture compositions or source replicate groups. These comparisons remain within one site-year and do not establish external environmental validity.

German source harmonization preserves the original site-table precision and follows the documented source treatment coding, while excluding drought-suffixed WF references from unsuffixed NF/WF pairs. One nonnumeric severity token remains unavailable. Treatment means retain yield and severity replicate counts separately; missing protected severity is not imputed. The primary endpoint population contains 3,264 comparisons across 16 site-years and five locations. The subset with observed severity on both management groups contains 954 comparisons across seven site-years and four locations. Common-population benchmarks accompany each comparison. The primary and temporal comparisons give site-years equal weight; geographic transfer gives locations equal weight. Site-specific rating comparability and incomplete assessment timing remain limitations.

The complete source inventory includes raw field observations, regional survey aggregates, published model-adjusted means, fitted curves and model inputs. Fitted response grids, duplicate archive copies and published parameter tables do not increase observational sample size. Older Nordic/Baltic yield and weather-derived recommendation tables do not provide

observed severity–yield pairs. New data-source identities, eligibility, fixed protocols, held-out predictions and independent numerical checks are retained with the current source data.

Weather and wheat-phenology contributions

The separate decomposition uses four historical–future combinations of disease weather and wheat-phenology trajectories in the 64-draw sample. Twenty pairs exceeding the accepted temperature range in a recombined trajectory are excluded. The two Shapley contributions share the interaction equally and sum to the net change on the same valid pairs. Phenology offsets use the identical pair set in numerator and denominator. The three-model ensemble offset is a ratio of mean contributions.

The main maps and regional estimates describe the completed full-grid experiment. Decomposition and structural sensitivity describe the separate sampled analysis. Field figures describe prediction errors on retrospective observations, and published yield figures describe treatment-mean contrasts. These datasets support different quantities and do not define a single overall model-accuracy percentage.

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Table S1a | German station calibration event-date evaluation. All process models use identical station/sowing-cycle/stage matches. Coverage includes observed events lacking shared predictions; Missing n counts unavailable predictions for the individual model. The distinct own-matched denominator is retained in Source Data. The station split does not withhold calendar years. Errors describe realized-weather, known-sowing hindcasts.

Model	BBCH	Matched n	Observed n	Coverage (%)	MAE (d)	RMSE (d)	Bias (d)	Missing n
GDD	10	5577	5699	97.9	5.35	8.99	1.26	122
GDD	31	5159	5159	100.0	17.22	24.80	-3.39	0
GDD	51	5352	5352	100.0	10.38	14.14	-0.96	0
GDD	85	1969	1969	100.0	10.46	13.88	-1.13	0
GDD	overall	18057	18179	99.3	10.79	16.76	-0.99	122
T-P	10	5577	5699	97.9	5.35	8.99	1.26	122
T-P	31	5159	5159	100.0	11.23	15.39	-0.50	0
T-P	51	5352	5352	100.0	6.03	8.23	0.55	0
T-P	85	1969	1969	100.0	7.07	9.26	0.13	0
T-P	overall	18057	18179	99.3	7.42	11.05	0.42	122
T-P-V	10	5577	5699	97.9	5.35	8.99	1.26	122
T-P-V	31	5159	5159	100.0	10.17	13.37	0.09	0
T-P-V	51	5352	5352	100.0	5.61	7.65	0.74	0
T-P-V	85	1969	1969	100.0	6.73	8.73	0.11	0
T-P-V	overall	18057	18179	99.3	6.96	10.08	0.65	122

Table S1b | German station validation event-date evaluation. All process models use identical station/sowing-cycle/stage matches. Coverage includes observed events lacking shared predictions; Missing n counts unavailable predictions for the individual model. The distinct own-matched denominator is retained in Source Data. The station split does not withhold calendar years. Errors describe realized-weather, known-sowing hindcasts.

Model	BBCH	Matched n	Observed n	Coverage (%)	MAE (d)	RMSE (d)	Bias (d)	Missing n
GDD	10	12653	13931	90.8	5.97	10.54	1.93	1278
GDD	31	11259	11699	96.2	18.99	27.27	-4.02	436
GDD	51	11636	12201	95.4	11.61	15.74	-0.48	554
GDD	85	2585	2706	95.5	11.96	15.87	-2.65	118
GDD	overall	38133	40537	94.1	11.94	18.68	-0.87	2386
T-P	10	12653	13931	90.8	5.97	10.54	1.93	1278
T-P	31	11259	11699	96.2	12.85	17.59	-1.27	436
T-P	51	11636	12201	95.4	7.08	9.74	0.53	552
T-P	85	2585	2706	95.5	7.99	10.53	-0.81	118
T-P	overall	38133	40537	94.1	8.48	12.83	0.37	2384
T-P-V	10	12653	13931	90.8	5.97	10.54	1.93	1278
T-P-V	31	11259	11699	96.2	11.63	15.32	-0.51	439
T-P-V	51	11636	12201	95.4	6.61	9.14	0.73	550
T-P-V	85	2585	2706	95.5	7.47	9.76	-0.44	118
T-P-V	overall	38133	40537	94.1	7.94	11.75	0.68	2385

Table S1c | German station testing event-date evaluation. All process models use identical station/sowing-cycle/stage matches. Coverage includes observed events lacking shared predictions; Missing n counts unavailable predictions for the individual model. The distinct own-matched denominator is retained in Source Data. The station split does not withhold calendar years. Errors describe realized-weather, known-sowing hindcasts.

Model	BBCH	Matched n	Observed n	Coverage (%)	MAE (d)	RMSE (d)	Bias (d)	Missing n
GDD	10	13471	14665	91.9	6.03	10.28	1.90	1194
GDD	31	12109	12539	96.6	19.10	27.23	-4.45	425
GDD	51	12541	13094	95.8	11.84	15.96	-0.86	531
GDD	85	2757	2879	95.8	11.75	15.20	-2.28	117
GDD	overall	40878	43177	94.7	12.07	18.66	-1.11	2267
T-P	10	13471	14665	91.9	6.03	10.28	1.90	1194
T-P	31	12109	12539	96.6	12.79	17.48	-1.48	426
T-P	51	12541	13094	95.8	7.29	10.11	0.30	544
T-P	85	2757	2879	95.8	7.85	10.17	-0.50	120
T-P	overall	40878	43177	94.7	8.54	12.79	0.25	2284
T-P-V	10	13471	14665	91.9	6.03	10.28	1.90	1194
T-P-V	31	12109	12539	96.6	11.59	15.27	-0.84	424
T-P-V	51	12541	13094	95.8	6.85	9.54	0.47	541
T-P-V	85	2757	2879	95.8	7.44	9.61	-0.27	120
T-P-V	overall	40878	43177	94.7	8.02	11.75	0.50	2279

Table S2 | Pooled field developmental-stage observation coverage and interval errors. Calibration and evaluation remain separate; eligible evaluation records from both archives are combined. Mean excess uses equal coordinate-year and field weighting on genuinely two-sided intervals. Unavailable intervals remain distinct from zero prediction error.

Cohort	BBCH	Fields	Two-sided n	Left n	Right n	Unavailable n	Mean excess (days)
Calibration	10	27	0	27	0	0	Unavailable
Calibration	31	27	0	27	0	0	Unavailable
Calibration	32	27	1	26	0	0	0.00
Calibration	33	27	5	22	0	0	0.00
Calibration	37	27	6	21	0	0	0.17
Calibration	39	27	9	18	0	0	1.33
Calibration	51	27	20	6	1	0	1.60
Calibration	65	27	20	4	3	0	1.47
Calibration	85	27	0	0	27	0	Unavailable
Field evaluation	10	161	0	161	0	0	Unavailable
Field evaluation	31	161	3	158	0	0	6.33
Field evaluation	32	161	8	153	0	0	0.75
Field evaluation	33	161	14	147	0	0	1.29
Field evaluation	37	161	22	137	2	0	1.95
Field evaluation	39	161	34	125	2	0	1.21
Field evaluation	51	161	73	81	7	0	1.24
Field evaluation	65	161	100	44	17	0	2.08
Field evaluation	85	158	5	0	153	0	10.50

Table S3 | Pooled disease evaluation and paired benchmark comparisons. Onset values are days outside two-sided symptom intervals; severity values are percentage points and detection values are percentages. Calibration records are excluded. Scores preserve the coordinate-year, field, source-leaf and assessment hierarchy. Intervals use 20,000 paired coordinate-year bootstrap resamples conditional on fixed predictions; source measurement conventions remain explicit in the underlying data.

Endpoint	Score	Estimate	95% interval	Records	Field seasons	Coordinate-years
symptom onset	model	5.322	3.734 to 7.087	136	78	62
symptom onset	benchmark	2.785	1.748 to 3.988	136	78	62
symptom onset	difference	2.536	1.154 to 3.925	136	78	62
severity	model rmse	31.193	28.473 to 33.876	2033	188	140
severity	model mae	20.088	17.892 to 22.356	2033	188	140
severity	model bias	-17.565	-20.220 to -14.941	2033	188	140
severity	benchmark rmse	29.834	27.514 to 32.099	2033	188	140
severity	benchmark mae	21.412	19.734 to 23.145	2033	188	140
severity	benchmark bias	-5.369	-8.121 to -2.701	2033	188	140
severity	rmse difference	1.359	-0.204 to 2.823	2033	188	140
severity	mae difference	-1.324	-2.788 to 0.127	2033	188	140
severity	bias difference	-12.196	-13.783 to -10.479	2033	188	140
severity	model r2	-0.125	Not estimated	2033	188	140
severity	benchmark r2	-0.029	Not estimated	2033	188	140
symptom detection	model sensitivity	97.156	95.925 to 98.179	2033	188	140
symptom detection	model specificity	45.065	35.780 to 55.652	2033	188	140
symptom detection	model balanced accuracy	71.110	66.607 to 76.236	2033	188	140
symptom detection	benchmark sensitivity	85.581	82.766 to 88.270	2033	188	140
symptom detection	benchmark specificity	72.014	62.312 to 82.501	2033	188	140
symptom detection	benchmark balanced accuracy	78.798	74.034 to 83.982	2033	188	140
symptom detection	sensitivity difference	11.575	9.048 to 14.205	2033	188	140
symptom detection	specificity difference	-26.950	-35.157 to -19.377	2033	188	140
symptom detection	balanced accuracy difference	-7.687	-11.888 to -3.804	2033	188	140

Table S4 | Fully nested selection re-estimates added stage thresholds, flowering proxy and weather preprocessing within each original-calibration training fold. Disease and phenology settings reproduce the original choices; three disease candidates share the nested minimum. Exact-minimum values and bracket counts show remaining threshold identification.

Model / stage	Selected threshold or setting	Nested loss / threshold range	Ties or exact minima	Original result	Original loss	Bracketed evidence
overwinter source model	a0.001_b10_L20_s120_i0.1	1.000	3	a0.001_b10_L20_s120_i0.1	1.000	8 / 6
phenology only	phenology_s80_L20	0.167	1	phenology_s80_L20	0.167	8 / 6
fold 0 BBCH 32	633.39	596.73 to 671.16	68	Censored-loss fit	—	1 genuinely two-sided fields
fold 0 BBCH 33	653.39	604.51 to 702.27	89	Censored-loss fit	—	3 genuinely two-sided fields
fold 0 BBCH 37	702.27	701.16 to 703.38	3	Censored-loss fit	—	4 genuinely two-sided fields
fold 0 BBCH 39	807.80	804.47 to 811.13	7	Censored-loss fit	—	6 genuinely two-sided fields
fold 1 BBCH 32	606.73	543.41 to 671.16	116	Censored-loss fit	—	0 genuinely two-sided fields
fold 1 BBCH 33	672.27	627.84 to 717.82	82	Censored-loss fit	—	4 genuinely two-sided fields
fold 1 BBCH 37	703.38	686.71 to 720.04	31	Censored-loss fit	—	5 genuinely two-sided fields
fold 1 BBCH 39	873.34	871.12 to 897.78	5	Censored-loss fit	—	7 genuinely two-sided fields
fold 2 BBCH 32	638.95	596.73 to 680.05	76	Censored-loss fit	—	1 genuinely two-sided fields
fold 2 BBCH 33	664.50	627.84 to 702.27	68	Censored-loss fit	—	3 genuinely two-sided fields
fold 2 BBCH 37	702.27	701.16 to 703.38	3	Censored-loss fit	—	3 genuinely two-sided fields
fold 2 BBCH 39	807.80	805.58 to 811.13	6	Censored-loss fit	—	5 genuinely two-sided fields

Table S5a | Published STB yield contrasts from Parker et al. (2004), Table 4. SX denotes Starcross and RM Rosemaund. Values are protected-minus-unprotected grain yield, t ha⁻¹. All 117 reported contrasts and five negative values are retained; dashes identify 33 unreported combinations. These are treatment contrasts from three-replicate experiments rather than raw plot observations.

Cultivar	SX95	SX96	SX97	RM95	RM96	RM97
Admiral	0.58	0.33	2.74	1.10	1.35	3.90
Andante	0.01	1.04	1.50	0.67	1.56	2.25
Avalon	-0.36	1.25	0.31	0.35	1.20	2.74
Beaver	0.31	0.87	1.85	0.64	1.15	3.69
Brigadier	0.74	1.65	3.87	0.91	2.11	4.50
Cadenza	-0.04	1.29	1.98	0.58	2.75	3.99
Estica	0.95	1.13	—	0.25	1.35	3.09
Flame	0.33	0.63	—	0.52	1.00	2.45
Galahad	-0.06	1.45	1.68	0.24	1.34	3.34
Haven	0.10	0.68	2.21	0.93	1.74	4.03
Hereward	0.59	1.04	1.90	0.61	1.64	2.52
Hornet	-0.23	—	—	0.43	—	—
Hunter	0.06	1.87	1.38	0.68	1.63	3.76
Hussar	0.67	1.87	2.80	0.89	1.54	4.03
Longbow	—	1.66	—	—	2.66	—
Lynx	-0.15	—	1.86	0.28	—	—
Mercia	0.69	1.58	2.21	1.08	2.49	2.82
Norman	0.46	2.06	1.94	0.91	2.07	4.01
Pastiche	—	1.07	—	—	1.10	1.14
Rialto	0.24	1.39	1.53	0.58	1.61	2.96
Riband	1.12	1.97	4.36	1.11	2.66	3.74
Ritmo	—	—	2.24	—	—	—
Soissons	—	—	0.31	—	—	—
Spark	—	—	1.20	—	—	—
Zodiac	0.45	1.40	—	1.25	1.59	—

Table S5b | Published top-three-leaf yield sensitivities from Parker et al. (2004), Table 5. All coefficients have units t ha^{-1} per GLAI-day. The population transfer centre is 0.0180; cultivar predictions span 0.0141–0.0207. Fixed estimates and their standard errors are reported for 21 cultivars. Random predictions share environments and are not independent replications.

Cultivar	Random prediction	Fixed estimate	Fixed SE
Mercia	0.0141	0.0153	0.0031
Avalon	0.0150	0.0146	0.0034
Flame	0.0161	0.0141	0.0033
Hunter	0.0162	0.0157	0.0031
Galahad	0.0170	0.0172	0.0034
Hussar	0.0171	0.0164	0.0031
Hereward	0.0171	0.0176	0.0035
Andante	0.0173	0.0185	0.0035
Pastiche	0.0174	0.0150	0.0048
Soissons	0.0180	—	—
Spark	0.0180	—	—
Ritmo	0.0180	—	—
Hornet	0.0180	—	—
Zodiac	0.0182	0.0174	0.0036
Estica	0.0184	0.0215	0.0036
Longbow	0.0185	0.0224	0.0042
Riband	0.0186	0.0225	0.0033
Cadenza	0.0191	0.0194	0.0033
Admiral	0.0192	0.0197	0.0034
Lynx	0.0192	0.0184	0.0038
Norman	0.0192	0.0220	0.0034
Beaver	0.0194	0.0189	0.0033
Haven	0.0201	0.0190	0.0034
Rialto	0.0203	0.0213	0.0033
Brigadier	0.0207	0.0206	0.0032

Table S6a | Foulkes et al. (2006), Table 2: paired mixed-model genotype-by-fungicide predicted means. HAD integrates the top five leaves from anthesis to complete canopy senescence, in GLAI-days. Yield is t ha⁻¹ at 85% dry matter. All eight contrasts are retained: two STB-only, two yellow-rust-only and four mixed-exposure means. The four site-years have unbalanced genotype and disease coverage. This broader canopy-yield evidence is not STB-specific epidemic validation. Average genotype-by-fungicide SED is 12.9 GLAI-days and 0.52 t ha⁻¹; these are not row-specific standard errors.

Genotype	Disease exposure	Protected HAD	Affected HAD	Protected yield	Affected yield
Mercia Rht-D1b	STB	129.2	77.5	6.49	5.20
Mercia Rht-D1a	STB	136.0	69.9	6.60	5.32
M. Huntsman Rht-D1b	Yellow rust	118.9	48.3	6.49	4.04
M. Huntsman Rht-D1a	Yellow rust	112.1	42.4	6.62	4.60
Hobbit awned	STB + yellow rust	118.9	60.6	7.04	4.80
Hobbit unawned	STB + yellow rust	114.6	64.1	6.45	5.08
Weston 1BL.1RS	STB + yellow rust	123.3	60.8	8.04	5.47
Chaucer 1BL.1BS	STB + yellow rust	112.7	59.3	7.69	5.26

Table S6b | Derived published contrasts and leave-one-genetic-background-out predictions. All eight disease-labelled contrasts are included. Each coefficient uses only the other four backgrounds. Relative losses divide protected-minus-affected differences by protected means; absolute response uses top-five HAD loss. Mercia, M. Huntsman, Hobbit, Weston and Chaucer receive equal total weights. Predictions reconstruct aggregate treatment means, rather than current-model field yield.

Background/line	HAD loss (%)	Yield loss (%)	Held prediction (%)	Yield loss (t ha ⁻¹)	Held prediction (t ha ⁻¹)
Mercia Rht-D1b	40.02	19.88	24.10	1.290	1.930
Mercia Rht-D1a	48.60	19.39	29.28	1.280	2.468
M. Huntsman Rht-D1b	59.38	37.75	34.65	2.450	2.482
M. Huntsman Rht-D1a	62.18	30.51	36.29	2.020	2.450
Hobbit awned	49.03	31.82	28.30	2.240	2.006
Hobbit unawned	44.07	21.24	25.43	1.370	1.737
Weston 1BL.1RS	50.69	31.97	28.54	2.570	2.025
Chaucer 1BL.1BS	47.38	31.60	26.40	2.430	1.718

Table S6c | Group-balanced prediction metrics for eight published mean contrasts in five backgrounds. Relative MAE, RMSE and bias are percentage points; absolute metrics are t ha^{-1} . The full-data absolute coefficient is t ha^{-1} per top-five GLAI-day, and the relative coefficient is dimensionless. Full-data coefficients are descriptive fits; metric calculations use held-background predictions. No covariance-adjusted confidence intervals are identified.

Response	MAE	RMSE	Bias	R ²	Full-data coefficient
Absolute	0.541	0.617	-0.009	-0.432	0.03427
Relative	4.793	5.156	0.022	0.247	0.57670

Table S7 | Compatibility of the retained published crop-response evidence. Disease exposure, leaf scope, integration window, predictor scale and grain-yield definition differ. The Foulkes dataset includes yellow-rust and mixed-disease contrasts. Study-specific observations, temporal scopes, units and limitations are retained in Source Data.

Study	Disease	Leaf scope	Predictor / outcome	Evidence role
Parker 2004	STB	Top three	GLAI-days / t ha ⁻¹	Top-three transfer
Foulkes 2006	STB; yellow rust	Top five	GLAI-days / t ha ⁻¹	Aggregate group holdout

Table S8 | Treatment-response prediction under held-out trial and year transfer. Trial-held-out endpoint and common-date window models retain their respective 67- and 50-contrast populations across five trial environments. Upper-three-leaf year transfer fits 2017–2018 treatment contrasts and evaluates 12 contrasts in 11 fields in 2019. Each model is compared with its own training-mean prediction; the populations and disease-percentage predictors retain distinct definitions. These comparisons evaluate management-associated relative yield responses rather than cause-specific STB yield loss or measured healthy-area duration.

Domain	Model	Contrasts	Environments / fields	RMSE (pp)	Baseline (pp)	MSE skill
Endpoint	Linear	67	5	4.164	3.884	-0.149
Window integral	Linear	50	5	5.557	3.985	-0.945
Four diseases	Linear	131	8	122.770	6.460	-360.228
Endpoint	Exponential	67	5	4.170	3.884	-0.153
Window integral	Exponential	50	5	5.540	3.985	-0.933
Four diseases	Exponential	131	8	19.386	6.460	-8.007
2019 year transfer; upper three leaves	Linear	12	11	5.23	3.60	-1.107
2019 year transfer; upper three leaves	Exponential	12	11	5.04	3.60	-0.958

Table S9 | Full-grid European climate impacts by emissions pathway and harvest period. Entries use fixed harvested-area weights over the valid historical-future season pairs for each metric. Timing is conditional on detected symptoms; occurrence includes symptom-free complete seasons. Relative HAD loss is HAD loss divided by reference HAD. Normalized HAD loss is HAD loss divided by maximum reference upper-canopy LAI, with units of days. Estimated disease-related yield change uses the reference coefficient and nominal reference LAI.

SSP	Period	Δ normalized HAD loss (days)	Δ timing (days)	Δ frequency (pp)	Δ relative HAD loss (pp)	Estimated Δ yield (kg ha ⁻¹ per unit LAI)	Valid HAD area-time (%)
SSP126	2031-2060	+1.11	-1.62	-0.026	+3.54	-20.1	99.958
SSP126	2071-2100	+1.45	-1.71	-0.033	+4.39	-26.0	99.956
SSP245	2031-2060	+0.77	-0.96	-0.069	+2.46	-13.8	99.960
SSP245	2071-2100	+1.55	-1.94	-0.219	+4.90	-27.8	99.913
SSP585	2031-2060	+1.53	-1.81	-0.087	+4.77	-27.5	99.958
SSP585	2071-2100	+2.66	-3.46	-0.400	+8.04	-47.8	99.922

Table S10 | Sensitivity of late-century SSP5–8.5 model responses to fourteen prespecified settings. The diagnostic sample retains 64 spatial draws at 62 cells. Metric-specific common paired populations and coverage are retained. Climate-model ranges and spatial sampling error refer to this diagnostic sample separately from the full-grid analysis. Yield quantities are conditional canopy-conversion products.

Setting	Outcome	Mean change	Three-GCM range	Spatial MCSE	Common-pair coverage
baseline	Flag-leaf symptoms (days relative to anthesis)	-3.797	-6.323 to -1.888	0.454	88.3–95.3%
baseline	Normalized HAD loss (days)	+2.617	+1.283 to +4.392	0.327	100.0–100.0%
baseline	Conditional yield effect (t ha ⁻¹ per nominal LAI)	+0.047	+0.023 to +0.079	0.006	100.0–100.0%
decline end0 conversion0p5	Flag-leaf symptoms (days relative to anthesis)	-3.797	-6.323 to -1.888	0.454	88.3–95.3%
decline end0 conversion0p5	Normalized HAD loss (days)	+0.629	+0.308 to +1.068	0.075	100.0–100.0%
decline end0 conversion0p5	Conditional yield effect (t ha ⁻¹ per nominal LAI)	+0.011	+0.006 to +0.019	0.001	100.0–100.0%
detection0p0001	Flag-leaf symptoms (days relative to anthesis)	-3.800	-7.331 to -1.109	0.888	88.3–95.3%
detection0p0001	Normalized HAD loss (days)	+2.617	+1.283 to +4.392	0.327	100.0–100.0%
detection0p0001	Conditional yield effect (t ha ⁻¹ per nominal LAI)	+0.047	+0.023 to +0.079	0.006	100.0–100.0%
detection0p01	Flag-leaf symptoms (days relative to anthesis)	-3.993	-5.811 to -2.493	0.718	88.3–95.3%
detection0p01	Normalized HAD loss (days)	+2.620	+1.287 to +4.397	0.328	100.0–100.0%
detection0p01	Conditional yield effect (t ha ⁻¹ per nominal LAI)	+0.047	+0.023 to +0.079	0.006	100.0–100.0%
functional conversion0p5	Flag-leaf symptoms (days relative to anthesis)	-3.797	-6.323 to -1.888	0.454	88.3–95.3%
functional conversion0p5	Normalized HAD loss (days)	+1.308	+0.642 to +2.196	0.164	100.0–100.0%
functional conversion0p5	Conditional yield effect (t ha ⁻¹ per nominal LAI)	+0.024	+0.012 to +0.040	0.003	100.0–100.0%
functional conversion0p75	Flag-leaf symptoms (days relative to anthesis)	-3.797	-6.323 to -1.888	0.454	88.3–95.3%
functional conversion0p75	Normalized HAD loss (days)	+1.962	+0.962 to +3.294	0.245	100.0–100.0%
functional conversion0p75	Conditional yield effect (t ha ⁻¹ per nominal LAI)	+0.035	+0.017 to +0.059	0.004	100.0–100.0%
local source off	Flag-leaf symptoms (days relative to anthesis)	-4.291	-6.861 to -2.272	0.459	88.3–95.3%
local source off	Normalized HAD loss (days)	+2.317	+1.005 to +4.136	0.290	100.0–100.0%
local source off	Conditional yield effect (t ha ⁻¹ per nominal LAI)	+0.042	+0.018 to +0.074	0.005	100.0–100.0%
primary objective tied runnerup	Flag-leaf symptoms (days relative to anthesis)	-3.524	-5.728 to -1.871	0.500	88.3–95.3%

Setting	Outcome	Mean change	Three-GCM range	Spatial MCSE	Common-pair coverage
primary objective tied runnerup	Normalized HAD loss (days)	+2.582	+1.265 to +4.388	0.322	100.0-100.0%
primary objective tied runnerup	Conditional yield effect (t ha ⁻¹ per nominal LAI)	+0.046	+0.023 to +0.079	0.006	100.0-100.0%
rank spacing160	Flag-leaf symptoms (days relative to anthesis)	-3.762	-6.438 to -1.730	0.473	88.3-95.3%
rank spacing160	Normalized HAD loss (days)	+2.801	+1.389 to +4.673	0.351	100.0-100.0%
rank spacing160	Conditional yield effect (t ha ⁻¹ per nominal LAI)	+0.050	+0.025 to +0.084	0.006	100.0-100.0%
rank spacing80	Flag-leaf symptoms (days relative to anthesis)	-3.348	-5.685 to -1.612	0.491	88.3-95.3%
rank spacing80	Normalized HAD loss (days)	+1.643	+0.631 to +3.001	0.225	100.0-100.0%
rank spacing80	Conditional yield effect (t ha ⁻¹ per nominal LAI)	+0.030	+0.011 to +0.054	0.004	100.0-100.0%
reference decline end0	Flag-leaf symptoms (days relative to anthesis)	-3.797	-6.323 to -1.888	0.454	88.3-95.3%
reference decline end0	Normalized HAD loss (days)	+1.258	+0.617 to +2.136	0.150	100.0-100.0%
reference decline end0	Conditional yield effect (t ha ⁻¹ per nominal LAI)	+0.023	+0.011 to +0.038	0.003	100.0-100.0%
reference decline end0p5	Flag-leaf symptoms (days relative to anthesis)	-3.797	-6.323 to -1.888	0.454	88.3-95.3%
reference decline end0p5	Normalized HAD loss (days)	+1.938	+0.950 to +3.264	0.238	100.0-100.0%
reference decline end0p5	Conditional yield effect (t ha ⁻¹ per nominal LAI)	+0.035	+0.017 to +0.059	0.004	100.0-100.0%
stage profile early37	Flag-leaf symptoms (days relative to anthesis)	-5.464	-9.195 to -2.675	0.671	88.3-95.3%
stage profile early37	Normalized HAD loss (days)	+2.549	+1.151 to +4.443	0.319	100.0-100.0%
stage profile early37	Conditional yield effect (t ha ⁻¹ per nominal LAI)	+0.046	+0.021 to +0.080	0.006	100.0-100.0%
stage profile late37	Flag-leaf symptoms (days relative to anthesis)	-1.543	-2.802 to -0.729	0.286	88.3-95.3%
stage profile late37	Normalized HAD loss (days)	+1.807	+0.819 to +3.123	0.248	100.0-100.0%
stage profile late37	Conditional yield effect (t ha ⁻¹ per nominal LAI)	+0.033	+0.015 to +0.056	0.004	100.0-100.0%

Table S11 | Parameter and scenario inventory. Values retain their model, calibration or registered-sensitivity provenance; uncertainty describes identification or scenario status, not a probability distribution unless explicitly stated. The complete nine-column machine-readable inventory is supplied as Parameter_inventory.csv.

Symbol	Parameter	Value	Unit	Role	Uncertainty
G_P_on	photoperiod_onset_gdd	142.01	effective degC d	fixed	Fixed in current study; inherited calibration; no current parameter interval.
G_P_off	photoperiod_stop_gdd	1426.78	effective degC d	fixed	Fixed in current study; inherited calibration; no current parameter interval.
U_V_on	vernalization_onset_tpp	142.01	effective degC d	fixed	Fixed in current study; inherited calibration; no current parameter interval.
U_V_off	vernalization_stop_tpp	582.8	effective degC d	fixed	Fixed in current study; inherited calibration; no current parameter interval.
T_base	thermal_zero_bound	0	degC	fixed	Intrinsic retained donor assumption; no uncertainty distribution.
T_cap	thermal_plateau_start	20	degC	fixed	Intrinsic retained donor assumption; no uncertainty distribution.
T_decline	thermal_decline_start	30	degC	fixed	Intrinsic retained donor assumption; no uncertainty distribution.
T_supported_max	maximum_supported_daily_mean	40	degC	fixed	Intrinsic retained donor assumption; no uncertainty distribution.
b_hot	thermal_decline_slope	2	effective degC per degC	fixed	Intrinsic retained donor assumption; no uncertainty distribution.
delta_amplitude	solar_declination_amplitude	0.4093	radian	fixed	Intrinsic retained donor assumption; no uncertainty distribution.
j_equ	solar_phase_day	81	day of year	fixed	Intrinsic retained donor assumption; no uncertainty distribution.
n_solar	astronomical_period	365	d	fixed	Intrinsic retained donor assumption; no uncertainty distribution.
ell_ref	photoperiod_reference_daylength	16	h	fixed	Intrinsic retained donor assumption; no uncertainty distribution.
b_P	photoperiod_response_slope	0.09	h ⁻¹	fixed	Intrinsic retained donor assumption; no uncertainty distribution.
v_knots_T	vernalization_temperature_knots	[-4, 0, 10, 16]	degC	fixed	Intrinsic retained donor assumption; no uncertainty distribution.

Symbol	Parameter	Value	Unit	Role	Uncertainty
v_knots_R	vernalization_response_at_knots	[0, 1, 1, 0]	effective vernalization d per d	fixed	Intrinsic retained donor assumption; no uncertainty distribution.
V_early	early_vernalization_hot_loss_bound	10	effective vernalization d	fixed	Intrinsic retained donor assumption; no uncertainty distribution.
T_devern	maximum_temperature_hot_loss_bound	30	degC	fixed	Intrinsic retained donor assumption; no uncertainty distribution.
b_devern	hot_loss_slope	0.5	effective vernalization d per degC per d	fixed	Intrinsic retained donor assumption; no uncertainty distribution.
V_complete	vernalization_response_saturation	40	effective vernalization d	fixed	Intrinsic retained donor assumption; no uncertainty distribution.
f_V_min	minimum_vernalization_multiplier	0.3	1	fixed	Intrinsic retained donor assumption; no uncertainty distribution.
f_V_span	vernalization_multiplier_increment	0.7	1	fixed	Intrinsic retained donor assumption; no uncertainty distribution.
q	development_multiplier	1	1	fixed	Retained q=1; no rescaling in current runtime.
C_10	BBCH10_threshold	142.01	effective degC d	fixed	Fixed in current study; inherited calibration; no current parameter interval.
C_31	BBCH31_threshold	542.3	effective degC d	fixed	Fixed in current study; inherited calibration; no current parameter interval.
C_32	BBCH32_threshold	633.39175	effective degC d	fitted	0-d loss tolerance 59 6.732875-671.1615; 1-d loss tolerance 54 3.410875-772.251125; 2-d loss tolerance 543.410875-830.016625; weak identification; not confidence intervals.
C_33	BBCH33_threshold	664.4962499999999	effective degC d	fitted	0-d loss tolerance 62 7.837375-702.266; 1-d loss tolerance 54 4.52175-834.460125; 2-d loss tolerance 5 44.52175-896.669125; weak identification; not confidence intervals.
C_37	BBCH37_threshold	702.266	effective degC d	fitted	0-d loss tolerance 70 1.155125-703.376875; 1-d loss tolerance 568.961-907.777875; 2-d loss tolerance 5 45.632625-979.98475; weak identification; not confidence intervals.

Symbol	Parameter	Value	Unit	Role	Uncertainty
C_39	BBCH39_threshold	807.799125	effective degC d	fitted	0-d loss tolerance 80 5.577375-811.13175 ; 1-d loss tolerance 6 35.6135-985.539125 ; 2-d loss tolerance 5 85.624125-985.539125; weak identification; not confidence intervals.
C_51	BBCH51_threshold	986.65	effective degC d	fixed	Fixed in current study; inherited calibration; no current parameter interval.
C_65	BBCH65_threshold	1165.925558	effective degC d	fitted	Inherited stage-only estimate; exact minimizing range 11 52.462482-1177.144788; 1-d loss tolerance 1069.44018-1264.654782; not confidence intervals.
C_85	BBCH85_threshold	1760.39	effective degC d	fixed	Fixed in current study; inherited calibration; no current parameter interval.
f_65	flowering_threshold_fraction	0.2317	1	fitted	Stage-only grid [0.02,0.60]; exact minimum fractions [0.2143,0.2462]; not a confidence interval.
Delta	rank_spacing_units	120.0	effective degC d	fitted	Selected from 80,120,160 with disease signs/timing; not leaf-count or measured-phyllchron calibration.
alpha	primary_scale	0.001	d ⁻¹ per relative-source unit	fitted	Grid 0.0001,0.001,0.01,0.1; source-scale dependent; no confidence interval.
beta	secondary_scale	10.0	d ⁻¹	fitted	Grid 0.1,1,10; selected upper grid boundary; weak identification; no confidence interval.
tau_L	latent_reference_days	20.0	reference d at 18 degC	fitted	Grid 20,30 d; effective latent-chain mean; finite-substep timing differs from continuous-time mean.
m	latent_stages	3	count	fixed	Fixed effective chain architecture; no uncertainty distribution.
tau_N	nonsporulating_reference_days	3.0	reference d at 18 degC	fixed	Effective assumption; not calibrated to disease magnitude.
tau_I	infectious_reference_days	21.0	reference d at 18 degC	fixed	Effective assumption; not calibrated to disease magnitude.

Symbol	Parameter	Value	Unit	Role	Uncertainty
tau_M	residue_maturation_reference_days	30.0	reference d at 18 degC and exposure=1	fixed	Effective source-competence assumption; no measured reproductive-maturation calibration.
tau_Q	residue_decay_reference_days	90.0	reference d at 18 degC	fixed	Effective common source-ageing assumption; same-fit half/double lifetime sensitivities.
f_0	initial_ready_fraction	0.5	1	fixed	Initial relative-source partition assumption; no measured readiness.
lambda	rank_distance_scale	2.0	rank intervals	fixed	Effective canopy geometry assumption; no calibrated physical transport distance.
p_star	rain_scale_mm	2.0	mm	fixed	Effective splash-response assumption; no uncertainty distribution.
omega	local_airborne_fraction	0.1	1	fixed	Relative local establishment mixture; not a measured spore-source share.
kappa	contact_fraction	0.2	1	fixed	Effective neighboring-leaf contact multiplier; contact-off structural sensitivity.
M	constant_imported_pressure	0.1	relative-source units	fitted	Grid 0,0.1; no observed imported source; source-scale dependent; no confidence interval.
r_star	rain_rate_mm_hour	0.45	mm h ⁻¹	fitted	Training-only meteorological median; 2695 eligible rain days at 6432 unique location-days; no sampling interval supplied.
J_policy	juvenile_policy	"handover_31_39"	policy	scenario	Primary handover_31_39; persistent alternative tested without fitting new ecological rates.
D_J	juvenile_renewal_degree_day_scale	300	positive degC d	fixed	Retained juvenile renewal assumption; no measured turnover calibration.
T_ref	disease_thermal_reference	18	degC	fixed	Defines relative stage/source clock; no fitted temperature-response uncertainty.
H_RH	humid_bin_cutoff	90	percent RH	fixed	Daily exposure reconstruction assumption; no measured canopy-wetness error distribution.

Symbol	Parameter	Value	Unit	Role	Uncertainty
eps_RH	humidity_cutoff_tolerance	1e-07	percentage points	fixed	Daily exposure reconstruction assumption; no measured canopy-wetness error distribution.
n_bin	synthetic_hourly_bins	24	count	fixed	Daily exposure reconstruction assumption; no measured canopy-wetness error distribution.
phi_bin	hourly_bin_phase_offset	0.5	h	fixed	Daily exposure reconstruction assumption; no measured canopy-wetness error distribution.
b_sat	saturation_exponent_coefficient	17.625	1	fixed	Daily exposure reconstruction assumption; no measured canopy-wetness error distribution.
T_sat	saturation_temperature_constant	243.04	degC	fixed	Daily exposure reconstruction assumption; no measured canopy-wetness error distribution.
n_bisect	humidity_bisection_iterations	30	count	fixed	Daily exposure reconstruction assumption; no measured canopy-wetness error distribution.
T_exp	exposure_temperature_optimum	18	degC	fixed	Daily exposure reconstruction assumption; no measured canopy-wetness error distribution.
sigma_exp	exposure_temperature_width	8	degC	fixed	Daily exposure reconstruction assumption; no measured canopy-wetness error distribution.
Q_0	initial_local_source	1.0	relative-source units	scenario	Conditional unit initial source; local-source-off intervention does not identify causal source effects.
Delta_t_req	requested_time_step	0.25	d	fixed	Frozen numerical setting; convergence checks assess numerical behavior rather than biological uncertainty.
n_sub	daily_substeps	4	count	fixed	$n = \text{ceil}(1/\text{requested_step})$; current effective substep is 0.25 d.
epsilon	detection_fraction	0.001	normalized tissue fraction	scenario	Fixed detection operator; post-fit cutoffs 0.0001 and 0.01; not biological infection threshold.

Symbol	Parameter	Value	Unit	Role	Uncertainty
D_miss	missing_positive_minimum_penalty	30	d	fixed	Scoring convention; missing-positive loss =max(30,forcing_end+1-upper_bound).
n_slot	leaf_slots	8	count	fixed	F1-F7 plus aggregate juvenile; does not estimate actual main-stem leaf number.
needs_V	vernalization_required	true	boolean	scenario	Winter-wheat requirement assumed; vernalization-off is a structural sensitivity.
L_star	standardized_upper3_lai	1.0	m ² lamina m ⁻² ground	scenario	Nominal scenario maximum; actual disease-free reference area and natural senescence unmeasured.
b_1	yield_transfer_slope	0.0141	t ha ⁻¹ per GLAI d	scenario	Published cultivar-range scenario; not confidence or prediction interval; no local yield calibration.
b_2	yield_transfer_slope	0.018	t ha ⁻¹ per GLAI d	scenario	Published cultivar-range scenario; not confidence or prediction interval; no local yield calibration.
b_3	yield_transfer_slope	0.0207	t ha ⁻¹ per GLAI d	scenario	Published cultivar-range scenario; not confidence or prediction interval; no local yield calibration.
Y_ref	reference_yield_t_ha	—	t ha ⁻¹	scenario	Absent by default; independently supplied positive value required for percentage diagnostic; not inferred.
f_proxy	functional_loss_conversion	"N+I+D"	normalized functional fraction	scenario	Unmeasured model proxy; latent tissue excluded; field/public runtime integrates pre-detection values.
gate_climate	climate_functional_detection_gate	"zero until first symptom fraction >= 0.001"	policy	scenario	Climate reporting convention only; differs from current field/public-runtime functional postprocessing.
W_window	yield_window	"inclusive first BBCH65 to first BBCH85"	calendar window	scenario	Development-window adaptation; GS31-85 sensitivity separately reported; no measured physiological yield window.
H	spatial_strata	16	count	fixed	Geographical stratification conditional on area/calendar registry.

Symbol	Parameter	Value	Unit	Role	Uncertainty
n_h	draws_per_stratum	4	count	fixed	Independent area-proportional replacement draws; MCSE separately estimated.
seed_spatial	sampling_seed	20261005	integer	fixed	Exact draw identities supplied; 64 draws at 62 unique cells.
phi_breaks	latitude_strata_breaks	[34, 43, 51, 59, 68]	degrees north	fixed	Design constants; no uncertainty distribution.
lon_breaks	longitude_strata_breaks	[-10, 8, 25, 42, 60]	degrees east	fixed	Design constants; no uncertainty distribution.
n_year	period_year_count	30	yr	fixed	Historical 1991–2020; future 2031–2060 and 2071–2100; relative-year pairing.
n_GCM	climate_model_count	3	count	scenario	ACCESS-CM2, MPI-ESM1-2-HR, MRI-ESM2-0; range is not a confidence interval.
calendar_type	regional_crop_calendar_type	"winter wheat/rainfed on all-wheat registered area"	policy	scenario	Genetic wheat type not verified; calendar and area mismatch is a structural limitation.

Table S12a | Four-combination Shapley decomposition for GS65–85 upper-three-leaf HAD proxy loss under SSP5–8.5. Interaction is allocated equally between the two contributions; weather plus host equals the net. Twenty of 5,751 finite pairs are excluded because a hybrid active-day disease-weather mean exceeds 40 °C. The seasons meeting the weather input rules and all decomposition terms share the remaining 5,731 pairs. Contributions are model-response components, not physical causal effects or measured yield.

Component	Mean (HAD days)	GCM range	Spatial MCSE	Coverage	Draw-year pairs
Weather contribution	+9.004	+6.528 to +12.221	0.621	99.20–99.86%	1909–1912
Wheat-phenology contribution	-6.360	-7.774 to -5.244	0.377	99.20–99.86%	1909–1912
Net HAD change	+2.644	+1.285 to +4.447	0.329	99.20–99.86%	1909–1912
Weather × host interaction	-2.784	-3.119 to -2.372	0.300	99.20–99.86%	1909–1912

Table S12b | Host development offsets part of the weather contribution to modeled canopy damage. HAD changes are days per unit nominal reference LAI on the common seasons meeting the weather input rules. Host offset is $-100 \times \text{host/weather}$; the ensemble percentage is the ratio of the three-model mean contributions, not the mean of the model-specific percentages. The interaction is already allocated equally to the weather and host terms. Percentages are descriptive model ratios without probability intervals and do not measure the benefit of an adaptation intervention.

Climate model	Weather ΔHAD	Host ΔHAD	Net ΔHAD	Host offset (%)
ACCESS-CM2	+12.221	-7.774	+4.447	63.6
MPI-ESM1-2-HR	+8.261	-6.061	+2.200	73.4
MRI-ESM2-0	+6.528	-5.244	+1.285	80.3
Three-model ensemble	+9.004	-6.360	+2.644	70.6

Table S13a | Full-grid late-century environmental-region changes in normalized HAD loss (days). Historical and future means use identical valid season pairs. Model ranges span three deterministic climate-model estimates. Unassigned and outside-region groups remain separate from the eight named EEA regions. All wheat grids are simulated, so these estimates have no spatial sampling error.

Region	SSP	Mean change	Three-model range	Eligible cells	Reference area (Mha)	Valid area-time (%)
Alpine	SSP126	+1.599	+0.817 to +2.454	828	0.996	99.025
Alpine	SSP245	+1.410	-0.128 to +3.308	828	0.996	99.049
Alpine	SSP585	+2.279	+1.376 to +3.994	828	0.996	99.043
Atlantic	SSP126	+1.358	-5.265 to +5.703	1455	5.964	99.964
Atlantic	SSP245	+1.999	-1.832 to +6.573	1455	5.964	99.967
Atlantic	SSP585	+4.849	+0.583 to +8.233	1455	5.964	99.967
Black Sea	SSP126	+0.780	-1.038 to +2.379	24	0.119	100.000
Black Sea	SSP245	+0.419	-1.267 to +2.563	24	0.119	100.000
Black Sea	SSP585	-0.621	-2.024 to +0.529	24	0.119	100.000
Boreal	SSP126	+2.611	-0.399 to +5.535	2914	3.317	99.716
Boreal	SSP245	+3.774	+2.141 to +6.900	2914	3.317	99.750
Boreal	SSP585	+7.841	+5.062 to +12.517	2914	3.317	99.729
Continental	SSP126	+1.966	-0.135 to +3.365	5057	23.097	100.000
Continental	SSP245	+2.007	+0.828 to +4.106	5057	23.097	100.000
Continental	SSP585	+3.453	+2.019 to +5.538	5057	23.097	99.990
Europe	SSP126	+1.446	-0.428 to +2.688	14932	54.405	99.956
Europe	SSP245	+1.546	+0.472 to +3.599	14932	54.405	99.913
Europe	SSP585	+2.656	+1.232 to +4.570	14932	54.405	99.922
Mediterranean	SSP126	+0.199	-0.456 to +0.571	1535	3.663	100.000
Mediterranean	SSP245	+0.090	-1.300 to +2.531	1535	3.663	100.000
Mediterranean	SSP585	-1.401	-1.878 to -1.129	1535	3.663	99.996
Outside EEA regions	SSP126	+0.389	+0.069 to +0.932	76	0.085	100.000
Outside EEA regions	SSP245	+0.768	+0.253 to +1.717	76	0.085	99.829
Outside EEA regions	SSP585	+1.735	+0.874 to +2.890	76	0.085	99.731
Pannonian	SSP126	+2.417	+1.678 to +3.177	285	1.766	100.000
Pannonian	SSP245	+2.087	+0.676 to +4.500	285	1.766	100.000
Pannonian	SSP585	+3.335	+2.350 to +5.170	285	1.766	100.000
Steppic	SSP126	+0.620	+0.370 to +0.749	2105	14.932	99.989
Steppic	SSP245	+0.484	+0.049 to +1.064	2105	14.932	99.819
Steppic	SSP585	+0.365	-0.078 to +1.228	2105	14.932	99.852
Unassigned	SSP126	+1.112	-2.251 to +3.568	653	0.466	99.826
Unassigned	SSP245	+1.196	-1.190 to +4.361	653	0.466	99.837
Unassigned	SSP585	+2.648	+0.169 to +5.269	653	0.466	99.832

Table S13b | Full-grid late-century environmental-region changes in relative HAD loss (percentage points). Historical and future means use identical valid season pairs. Model ranges span three deterministic climate-model estimates. Unassigned and outside-region groups remain separate from the eight named EEA regions. All wheat grids are simulated, so these estimates have no spatial sampling error.

Region	SSP	Mean change	Three-model range	Eligible cells	Reference area (Mha)	Valid area-time (%)
Alpine	SSP126	+5.525	+3.193 to +8.683	828	0.996	99.025
Alpine	SSP245	+5.743	+0.551 to +11.942	828	0.996	99.049
Alpine	SSP585	+8.537	+5.874 to +13.801	828	0.996	99.043
Atlantic	SSP126	+4.256	-13.506 to +16.722	1455	5.964	99.964
Atlantic	SSP245	+6.632	-4.025 to +19.900	1455	5.964	99.967
Atlantic	SSP585	+15.349	+3.207 to +25.356	1455	5.964	99.967
Black Sea	SSP126	+1.959	-3.541 to +6.973	24	0.119	100.000
Black Sea	SSP245	+0.776	-3.909 to +6.286	24	0.119	100.000
Black Sea	SSP585	-3.295	-7.095 to -1.373	24	0.119	100.000
Boreal	SSP126	+8.272	-0.840 to +17.742	2914	3.317	99.716
Boreal	SSP245	+11.936	+6.470 to +22.201	2914	3.317	99.750
Boreal	SSP585	+23.990	+15.300 to +38.564	2914	3.317	99.729
Continental	SSP126	+5.893	-0.103 to +10.355	5057	23.097	100.000
Continental	SSP245	+6.361	+2.461 to +12.823	5057	23.097	100.000
Continental	SSP585	+10.489	+6.467 to +16.592	5057	23.097	99.990
Europe	SSP126	+4.387	-0.831 to +8.222	14932	54.405	99.956
Europe	SSP245	+4.898	+1.723 to +11.050	14932	54.405	99.913
Europe	SSP585	+8.037	+4.077 to +13.527	14932	54.405	99.922
Mediterranean	SSP126	+0.572	-1.226 to +1.703	1535	3.663	100.000
Mediterranean	SSP245	+0.341	-3.551 to +7.227	1535	3.663	100.000
Mediterranean	SSP585	-4.001	-5.268 to -3.336	1535	3.663	99.996
Outside EEA regions	SSP126	+1.544	+0.390 to +3.534	76	0.085	100.000
Outside EEA regions	SSP245	+2.812	+0.957 to +6.099	76	0.085	99.829
Outside EEA regions	SSP585	+5.830	+2.974 to +9.544	76	0.085	99.731
Pannonian	SSP126	+7.344	+5.225 to +9.014	285	1.766	100.000
Pannonian	SSP245	+6.550	+1.990 to +13.924	285	1.766	100.000
Pannonian	SSP585	+9.894	+6.960 to +15.285	285	1.766	100.000
Steppic	SSP126	+1.831	+1.050 to +2.298	2105	14.932	99.989
Steppic	SSP245	+1.334	+0.017 to +2.861	2105	14.932	99.819
Steppic	SSP585	+0.592	-0.418 to +2.504	2105	14.932	99.852
Unassigned	SSP126	+3.195	-5.573 to +10.097	653	0.466	99.826
Unassigned	SSP245	+3.515	-2.957 to +12.395	653	0.466	99.837
Unassigned	SSP585	+7.725	+0.914 to +14.958	653	0.466	99.832

Table S13c | Full-grid late-century environmental-region changes in symptom timing (days relative to flowering). Historical and future means use identical valid season pairs. Model ranges span three deterministic climate-model estimates. Unassigned and outside-region groups remain separate from the eight named EEA regions. All wheat grids are simulated, so these estimates have no spatial sampling error.

Region	SSP	Mean change	Three-model range	Eligible cells	Reference area (Mha)	Valid area-time (%)
Alpine	SSP126	-1.425	-1.978 to -0.702	828	0.996	98.640
Alpine	SSP245	-1.224	-2.955 to -0.296	828	0.996	98.386
Alpine	SSP585	-2.474	-4.359 to -1.297	828	0.996	98.699
Atlantic	SSP126	-0.111	-2.602 to +4.272	1455	5.964	99.964
Atlantic	SSP245	-1.101	-3.589 to +0.857	1455	5.964	99.967
Atlantic	SSP585	-3.233	-5.218 to -0.559	1455	5.964	99.967
Black Sea	SSP126	-1.621	-2.480 to -0.769	24	0.119	99.211
Black Sea	SSP245	-1.462	-4.376 to +0.366	24	0.119	99.894
Black Sea	SSP585	-0.843	-2.635 to +1.796	24	0.119	100.000
Boreal	SSP126	-2.815	-5.338 to -0.080	2914	3.317	99.715
Boreal	SSP245	-3.251	-5.841 to -1.425	2914	3.317	99.750
Boreal	SSP585	-7.434	-11.059 to -4.624	2914	3.317	99.729
Continental	SSP126	-2.183	-3.311 to -0.438	5057	23.097	99.814
Continental	SSP245	-2.248	-3.699 to -1.364	5057	23.097	99.881
Continental	SSP585	-4.409	-6.952 to -2.334	5057	23.097	99.845
Europe	SSP126	-1.708	-2.689 to -0.081	14932	54.405	99.125
Europe	SSP245	-1.941	-3.687 to -0.763	14932	54.405	98.996
Europe	SSP585	-3.463	-6.020 to -1.572	14932	54.405	98.655
Mediterranean	SSP126	+0.052	-1.785 to +1.191	1535	3.663	98.131
Mediterranean	SSP245	+1.408	-1.597 to +2.954	1535	3.663	94.600
Mediterranean	SSP585	+3.858	+3.054 to +4.872	1535	3.663	89.308
Outside EEA regions	SSP126	-2.032	-2.899 to -1.460	76	0.085	96.656
Outside EEA regions	SSP245	-2.765	-3.815 to -1.143	76	0.085	96.769
Outside EEA regions	SSP585	-4.621	-5.663 to -4.078	76	0.085	97.328
Pannonian	SSP126	-2.365	-3.826 to -1.502	285	1.766	99.989
Pannonian	SSP245	-1.233	-2.618 to -0.478	285	1.766	100.000
Pannonian	SSP585	-3.110	-4.688 to -1.522	285	1.766	100.000
Steppic	SSP126	-1.745	-2.124 to -1.357	2105	14.932	97.699
Steppic	SSP245	-2.451	-3.891 to -1.310	2105	14.932	98.009
Steppic	SSP585	-3.024	-6.374 to -1.216	2105	14.932	98.094
Unassigned	SSP126	-1.132	-2.744 to +0.985	653	0.466	99.795
Unassigned	SSP245	-1.628	-4.164 to +0.147	653	0.466	99.725
Unassigned	SSP585	-2.283	-4.063 to -0.990	653	0.466	99.569

Table S14 | Country-group climate impacts under late-century SSP5-8.5. Groups follow the dominant SPAM source-country assignment of quarter-degree wheat cells within the study domain. Values are exact harvested-area-weighted model scenario responses rather than observed national averages. Cross-border cells are not partitioned, and the Russian Federation and Turkey entries cover only represented domain cells. All SSP-specific estimates, climate-model ranges and valid area-time coverage remain in Source Data.

Country group	Eligible cells	Reference area (Mha)	Δ normalized HAD loss (days)	Δ timing (days)	Δ relative HAD loss (pp)	Estimated Δ yield (kg ha ⁻¹ per unit LAI)
Russian Federation	4001	19.723	+1.74	-4.19	+5.19	-31.3
Ukraine	1211	6.806	+0.18	-3.48	-0.56	-3.2
France	1083	5.015	+4.13	-2.42	+13.49	-74.3
Germany	714	2.971	+7.41	-5.68	+22.52	-133.3
Poland	643	2.409	+8.48	-7.35	+26.12	-152.6
Romania	437	2.155	+0.75	-0.93	+2.51	-13.5
Spain	920	1.988	-0.80	+4.05	-1.86	+14.4
Italy	578	1.730	-2.02	+2.94	-6.99	+36.3
United Kingdom	482	1.674	+5.21	-3.75	+16.42	-93.8
Bulgaria	200	1.204	-1.42	+1.02	-4.74	+25.5
Hungary	175	0.940	+3.43	-3.19	+10.25	-61.8
Lithuania	150	0.916	+10.43	-8.69	+31.19	-187.7
Czechia	161	0.815	+4.26	-4.71	+13.83	-76.7
Belarus	453	0.707	+3.78	-5.20	+11.93	-68.1
Serbia	153	0.601	+1.67	-1.77	+4.78	-30.0
Denmark	141	0.537	+10.09	-7.55	+29.67	-181.7
Latvia	169	0.510	+10.80	-8.97	+32.18	-194.4
Sweden	748	0.466	+6.62	-7.46	+20.95	-119.2
Slovakia	103	0.389	+4.68	-3.86	+14.97	-84.2
Turkey	54	0.364	-0.52	+0.42	-2.89	+9.4
Greece	305	0.359	-1.61	+3.24	-4.20	+28.9
Republic of Moldova	62	0.332	+0.26	-1.42	+0.56	-4.6
Austria	147	0.266	+1.52	-2.15	+8.03	-27.3
Belgium	57	0.204	+6.04	-3.89	+19.41	-108.8
Finland	781	0.203	+5.29	-6.87	+17.52	-95.2
Estonia	80	0.171	+10.88	-9.75	+32.95	-195.9
Kazakhstan	133	0.156	+0.66	-5.57	+1.91	-11.9
Croatia	114	0.143	+2.00	-1.55	+6.43	-36.0
Netherlands	58	0.110	+7.10	-4.69	+21.45	-127.8
Switzerland	29	0.086	+3.85	-2.69	+13.68	-69.3
Azerbaijan	18	0.072	+4.91	-8.31	+9.31	-88.4
Norway	248	0.071	+6.21	-5.95	+20.21	-111.7
North Macedonia	41	0.067	-0.04	+1.59	+0.00	+0.8
Bosnia and Herzegovina	49	0.064	+2.01	-1.20	+6.90	-36.2
Ireland	46	0.058	+3.58	-1.48	+14.01	-64.5
Albania	39	0.055	-0.88	+1.87	-2.44	+15.8
Slovenia	39	0.028	+2.94	-2.53	+9.19	-52.9
Portugal	103	0.027	-2.50	+3.80	-5.91	+44.9
Luxembourg	4	0.014	+5.60	-3.66	+18.59	-100.9
Montenegro	1	0.001	-0.10	-2.31	-5.84	+1.9
Georgia	2	0.000	-0.28	-1.01	+6.06	+5.0

Table S15 | Exact harvested-area shares by climate-model canopy-response sign. Shares use all cells with complete three-model estimates under each scenario–period comparison. Positive ensemble means and unanimous positive signs are distinct classifications. Mixed signs include at least one positive and one negative model change; any zero-only combinations remain in Source Data.

SSP	Period	Positive ensemble (%)	All models positive (%)	All models negative (%)	Mixed signs (%)	Complete-model area (%)
SSP126	2031-2060	95.83	51.21	0.91	47.88	99.984
SSP126	2071-2100	93.36	50.01	0.78	49.22	99.983
SSP245	2031-2060	71.53	21.04	9.80	69.16	99.984
SSP245	2071-2100	86.22	55.36	5.70	38.94	99.984
SSP585	2031-2060	89.07	50.23	4.52	45.25	99.985
SSP585	2071-2100	77.29	65.17	14.86	19.98	99.985

Table S16 | Full-grid wheat-land-use and environmental-region coverage. Areas sum the fixed SPAM2020 all-wheat harvested-area registry. Calendar eligibility uses the imposed GGCMI winter-wheat rainfed scenario. Environmental assignment follows cell centroids; missing and outside-region assignments remain explicit.

Environmental region	Land-use cells	Eligible calendars	Reference wheat area (Mha)
Alpine	828	828	0.9959
Atlantic	1457	1455	5.9642
Black Sea	24	24	0.1187
Boreal	2914	2914	3.3170
Continental	5057	5057	23.0971
Mediterranean	1535	1535	3.6633
Outside EEA regions	76	76	0.0846
Pannonian	285	285	1.7656
Steppic	2108	2105	14.9322
Unassigned	657	653	0.4664

Table S17 | Additional field evidence for disease-related yield-response evaluation. The Tunisia/Nordic combined source inventory contains 423 assessments and 272 yield keys; excluding post-harvest measurements and one pooled-control alias leaves 412 assessments and 263 outcomes. Counts retain plot versus treatment-mean grain and do not represent independent environments. The German archive is separate from this combined inventory. None supplies compatible repeated leaf-area and harvested-yield measurements sufficient to validate the climate HAD conversion.

Source	Matched yield outcomes	Predictive comparisons	Environmental coverage	Measurement support
Nordic extension	181 eligible treatment means	62 treated-untreated contrasts	5 prediction trials; 11 inventory trials	Dated symptoms; leaf ranks unconfirmed in prediction trials
Tunisia	82 unprotected plots; 40 protected references	82 cross-year yields; 40 reference gaps	2 years at one site; references only in 2019	Flag and lower leaf on different dates; protected severity missing
Germany	Cultivar treatment means from original site files	3,264 matched protection responses	16 site-years; 5 locations	Whole-plot severity; assessment dates and leaf ranks unspecified

Table S18 | Nordic grouped yield-response comparisons. The target is 1 – untreated yield / treated yield; values are percentage points. All candidates use their own common-population training-mean benchmark. Whole trials retain shared controls and repeated disease assessments. Nested selection uses training trials only. Intervals resample complete trials conditional on fixed held-out predictions and exclude refitting uncertainty. The nonnegative stage row is a secondary diagnostic after inspection of the unconstrained stage-model slopes. All specified model scores remain in Source Data; fitted coefficients accompany the held-out prediction CSVs in the evidence archive.

Population / validation	Model	n / trials	RMSE (pp)	Mean benchmark (pp)	95% ΔRMSE (pp)
Endpoint / trial holdout	Training mean	62 / 5	5.02	5.02	+0.00 to +0.00
Endpoint / trial holdout	Severity, linear through origin	62 / 5	7.24	5.02	+0.00 to +4.01
Endpoint / trial holdout	Severity, exponential	62 / 5	7.17	5.02	-0.01 to +3.93
Endpoint / trial holdout	Severity + intercept	62 / 5	4.83	5.02	-0.60 to +0.64
Endpoint / trial holdout	Nested model selection	62 / 5	4.88	5.02	-0.33 to +0.00
Stage recorded / trial holdout	Training mean	51 / 4	5.79	5.79	+0.00 to +0.00
Stage recorded / trial holdout	Severity + intercept	51 / 4	5.82	5.79	-0.86 to +1.67
Stage recorded / trial holdout	Severity + assessment stage	51 / 4	3.99	5.79	-3.38 to +1.09
Stage recorded / trial holdout	Nested model selection	51 / 4	6.06	5.79	-0.01 to +1.10
Two assessments / trial holdout	Training mean	40 / 5	5.38	5.38	+0.00 to +0.00
Two assessments / trial holdout	Early + late severity, nonnegative	40 / 5	7.97	5.38	+1.11 to +4.18
Two assessments / trial holdout	Early + late severity	40 / 5	6.04	5.38	-0.26 to +2.26
Endpoint / 2024→2025	Training mean	38 / 3	5.06	5.06	Not estimated
Endpoint / 2024→2025	Severity, linear through origin	38 / 3	5.62	5.06	Not estimated
Endpoint / 2024→2025	Severity + intercept	38 / 3	5.04	5.06	Not estimated
Endpoint / 2024→2025	Training-selected model	38 / 3	5.06	5.06	Not estimated
Stage recorded / secondary	Stage response, nonnegative	51 / 4	7.55	5.79	+0.54 to +3.49

Table S19 | Tunisian leaf-rank models under grouped yield evaluation. Absolute-yield transfer gives the two years equal weight. Within-2019 response is 1 – unprotected yield / source-matched protected yield, with complete mixture compositions or source replicate groups excluded from fitting. The latter tests remain within one environment. Flag-leaf and lower-leaf predictors are observed incidence-weighted lesion percentages; protected severity is unavailable. Cultivar-composition predictors use four source cultivar proportions. The nonnegative model constrains higher damage to lower absolute yield or a larger relative protected-reference gap. One negative measured reference gap remains in all compatible comparisons.

Prediction task	Model	n / groups	RMSE	Mean benchmark	Units
Between years	Training mean	82 / 2	3.24	3.24	t ha ⁻¹
Between years	Cultivar composition	82 / 2	3.12	3.24	t ha ⁻¹
Between years	Flag-leaf severity	82 / 2	3.27	3.24	t ha ⁻¹
Between years	Lower-leaf severity	82 / 2	3.32	3.24	t ha ⁻¹
Between years	Both leaf severities	82 / 2	3.32	3.24	t ha ⁻¹
Between years	Composition + both leaves	82 / 2	3.28	3.24	t ha ⁻¹
Between years	Nonnegative damage	82 / 2	3.42	3.24	t ha ⁻¹
2019 mixture holdout	Training mean	40 / 10	10.06	10.06	percentage points
2019 mixture holdout	Cultivar composition	40 / 10	10.68	10.06	percentage points
2019 mixture holdout	Flag-leaf severity	40 / 10	9.92	10.06	percentage points
2019 mixture holdout	Lower-leaf severity	40 / 10	10.03	10.06	percentage points
2019 mixture holdout	Both leaf severities	40 / 10	10.12	10.06	percentage points
2019 mixture holdout	Composition + both leaves	40 / 10	10.48	10.06	percentage points
2019 mixture holdout	Nonnegative damage	40 / 10	10.06	10.06	percentage points
2019 replicate holdout	Training mean	40 / 4	10.01	10.01	percentage points
2019 replicate holdout	Cultivar composition	40 / 4	10.33	10.01	percentage points
2019 replicate holdout	Flag-leaf severity	40 / 4	9.76	10.01	percentage points
2019 replicate holdout	Lower-leaf severity	40 / 4	10.02	10.01	percentage points
2019 replicate holdout	Both leaf severities	40 / 4	9.99	10.01	percentage points
2019 replicate holdout	Composition + both leaves	40 / 4	10.38	10.01	percentage points
2019 replicate holdout	Nonnegative damage	40 / 4	10.01	10.01	percentage points

Table S20 | German protection-response models under spatial and temporal transfer. All comparisons use source original site files, matching cultivar, nitrogen and source-coded water regime. The response is 1 – unprotected treatment-mean yield / protected treatment-mean yield. Drought-suffixed references and nonpositive yields are excluded; missing disease measurements are not imputed. Site-year and temporal scores weight site-years equally; geographic transfer weights locations equally. The observed-reduction subset requires recorded severity in both treatment groups, with its own benchmark. Treatment means retain their different disease and yield replicate coverage. These protection responses concern multiple diseases and are not causal estimates of STB-specific loss.

Population / validation	Model	n / groups	RMSE (pp)	Mean benchmark (pp)
Unprotected endpoint / site-year holdout	Training mean	3264 / 16	17.57	17.57
Unprotected endpoint / site-year holdout	Nitrogen + water regime	3264 / 16	16.82	17.57
Unprotected endpoint / site-year holdout	Septoria severity	3264 / 16	17.66	17.57
Unprotected endpoint / site-year holdout	Severity + management	3264 / 16	16.92	17.57
Unprotected endpoint / site-year holdout	Nonnegative damage	3264 / 16	17.67	17.57
Unprotected endpoint / site holdout	Training mean	3264 / 5	18.48	18.48
Unprotected endpoint / site holdout	Nitrogen + water regime	3264 / 5	18.32	18.48
Unprotected endpoint / site holdout	Septoria severity	3264 / 5	18.51	18.48
Unprotected endpoint / site holdout	Severity + management	3264 / 5	18.48	18.48
Unprotected endpoint / site holdout	Nonnegative damage	3264 / 5	18.51	18.48
Unprotected endpoint / 2018–2019 transfer	Training mean	619 / 6	19.56	19.56
Unprotected endpoint / 2018–2019 transfer	Nitrogen + water regime	619 / 6	17.55	19.56
Unprotected endpoint / 2018–2019 transfer	Septoria severity	619 / 6	19.63	19.56
Unprotected endpoint / 2018–2019 transfer	Severity + management	619 / 6	17.51	19.56
Unprotected endpoint / 2018–2019 transfer	Nonnegative damage	619 / 6	19.65	19.56
Both severities observed / site-year holdout	Training mean	954 / 7	16.29	16.29
Both severities observed / site-year holdout	Septoria severity	954 / 7	16.47	16.29
Both severities observed / site-year holdout	Observed severity reduction	954 / 7	17.23	16.29
Both severities observed / site-year holdout	Nonnegative damage	954 / 7	17.42	16.29
Both severities observed / site holdout	Training mean	954 / 4	16.95	16.95
Both severities observed / site holdout	Septoria severity	954 / 4	22.62	16.95
Both severities observed / site holdout	Observed severity reduction	954 / 4	23.45	16.95
Both severities observed / site holdout	Nonnegative damage	954 / 4	24.51	16.95
Both severities observed / 2018–2019 transfer	Training mean	515 / 5	22.37	22.37
Both severities observed / 2018–2019 transfer	Septoria severity	515 / 5	22.91	22.37
Both severities observed / 2018–2019 transfer	Observed severity reduction	515 / 5	22.86	22.37

Population / validation	Model	n / groups	RMSE (pp)	Mean benchmark (pp)
Both severities observed / 2018-2019 transfer	Nonnegative damage	515 / 5	22.94	22.37

Table S21 | European production exposure across scenarios and periods. Fixed SPAM2020 production is located in cells with positive ensemble-mean changes or positive changes in all three climate models. These overlapping quantities are not additive and do not estimate production losses. Regional, fractional source-country and 27-of-30-pair sensitivity results are supplied in Source Data.

SSP	Period	Baseline (Mt)	Ensemble-increase exposure (Mt)	All-model increase exposure (Mt)	Unavailable (Mt)	Production-weighted HAD change (days)
SSP126	2031-2060	241.845	232.010	98.121	0.033	+1.145
SSP126	2071-2100	241.845	228.909	97.044	0.034	+1.604
SSP245	2031-2060	241.845	181.824	53.256	0.033	+1.000
SSP245	2071-2100	241.845	212.111	118.557	0.033	+1.784
SSP585	2031-2060	241.845	219.614	131.046	0.033	+1.836
SSP585	2071-2100	241.845	194.868	162.212	0.033	+3.356

Table S22 | Crop and symptom timing on identical complete season pairs. Reference and future periods are 1991–2020 and 2071–2100 under SSP5–8.5. All events require finite dates in both seasons. Event dates are elapsed calendar days after sowing; relative symptom timing and grain-fill elapsed duration are date differences. Fixed harvested-area and valid-pair weights precede equal climate-model averaging. All event differences therefore reconcile on the same population.

Event or interval	Reference (days)	Future (days)	Change (days)	Climate-model change range (days)
flowering days	255.604	228.289	-27.315	-39.366 to -17.165
grain fill elapsed days	33.223	33.472	+0.249	-0.491 to +1.192
soft dough days	288.827	261.761	-27.066	-38.174 to -17.656
symptom days	246.100	215.322	-30.778	-45.386 to -18.737
symptom relative flowering days	-9.504	-12.968	-3.463	-6.020 to -1.572

Table S23 | Decision relevance and unresolved evidence requirements. Production exposure supports conditional prioritization of observation and evaluation. The weather–phenology decomposition does not estimate management efficacy, and baseline production located in exposed cells does not measure lost tonnes.

Decision	Evidence available	Additional evidence required
Regional surveillance and field-trial coverage	Production exposure, projected damage direction and climate-model disagreement	Field verification; separate winter/spring and rainfed/irrigated wheat representation
Sowing-date or cultivar choice	Climate-driven timing shifts and model decomposition	Feasible management comparisons with grain yield, heat/water exposure and costs
Disease-attributable grain loss	Conditional canopy damage; limited treatment-mean yield evidence	Matched functional canopy and harvest observations; independent crop-growth and disease-loss evaluation
Avoided production loss or input-reduction targets	No validated intervention-effect estimate	Paired crop-growth scenarios with and without disease, field intervention trials and implementation costs

Table S24 | Measurement compatibility and evidence requirements for grain-loss and adaptation claims. Availability of observed grain yield supplies a prediction target; it does not establish a validated crop-disease-yield chain. Ordinal ratings remain source-specific categories. The inventory contains 948 additional unique plot harvests, including 724 Swiss plots and 224 French plots; no additional compatible functional-canopy series or identified STB-specific yield counterfactual was found. Source metadata, units, duplicate checks and raw-yield reconciliation are retained in Source Data and the public source subset.

Quantity	Available evidence	Missing evidence for the main inference
disease	Observed source-specific disease ratings and existing external disease/response benchmarks	Dated repeated quantitative leaf-area severity, leaf ranks and observation functions for canopy-linked validation
HAD	Conditional proxy diagnostics and archived physiological summaries only	Repeated dated functional green/reference LAI by leaf rank across anthesis to maturity, measured senescence and uncertainty
grain yield	948 unique observed plot harvests: 724 Swiss and 224 French; existing benchmarks retained	Matched canopy/phenology/weather/soil/management and independent yield predictions for end-to-end crop-model validation
grain loss	Existing treatment-associated yield responses; new sources lack an identified STB yield counterfactual	Comparable randomized/matched reference and exposed plots with measured disease in both, functional canopy, harvest and other stresses
tonnes	Conditional scenarios only; no validated regional grain-loss conversion	Validated absolute yield changes, crop area and production baselines, moisture consistency, spatial representativeness and uncertainty
adaptation	Recorded mixture composition and management; existing treatment-response comparisons	Replicated feasible intervention contrasts with matched physiology, yield, multiple environments and implementation/cost constraints

Table S25 | Exploratory Swiss harvest-yield prediction across sites and site-years. All six fixed models use the same 724 plot harvests across ten site-years at five sites, with each site-year receiving equal evaluation weight. Whole-site validation is primary; site-year validation is secondary. Encodings and fitting are restricted to each training fold. STB and other disease ratings remain ordinal categories; disease dates are unavailable. Cultivar identities can recur across folds. The protocol was specified after source inspection and before fitting, without external preregistration. Five-site descriptive resampling of the incremental STB comparison includes both improvement and deterioration. These retrospective associations do not identify STB-attributable yield loss or management efficacy.

Held-out group	Predictors	Yield RMSE (t ha ⁻¹)	Yield MAE (t ha ⁻¹)	MSE skill versus training mean
site	Cultivar / management	0.914	0.757	0.359
site	Cultivar / management + other diseases	0.951	0.757	0.306
site	Cultivar / management + other diseases + STB	0.932	0.743	0.334
site	Cultivar / management + STB	0.877	0.706	0.410
site	Ordinal STB	0.926	0.747	0.342
site	Training mean	1.142	0.928	0.000
site year	Cultivar / management	0.998	0.833	0.259
site year	Cultivar / management + other diseases	1.064	0.860	0.158
site year	Cultivar / management + other diseases + STB	1.005	0.802	0.249
site year	Cultivar / management + STB	0.915	0.738	0.377
site year	Ordinal STB	0.972	0.785	0.297
site year	Training mean	1.159	0.940	0.000

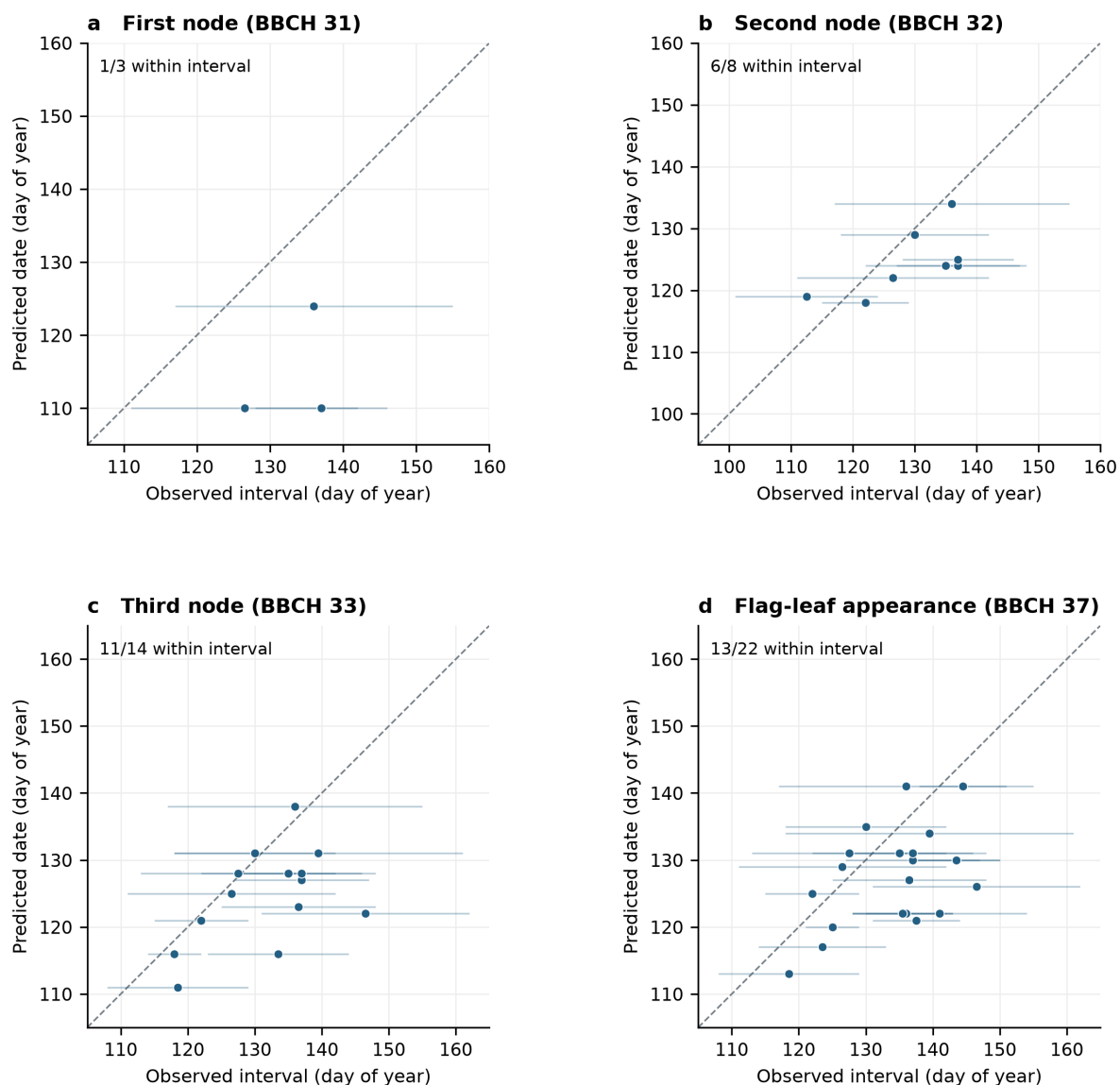


Figure S1 | Field evaluation of wheat phenological stage dates. Horizontal segments show observed two-sided stage intervals; their centers locate markers and are not treated as exact event dates. The dashed diagonal denotes equal dates, and an interval intersecting it contains the prediction. Counts give compatible intervals among all informative records. Observations from both field archives are pooled; calibration and evaluation remain separate. Intervals express observation resolution rather than statistical confidence.

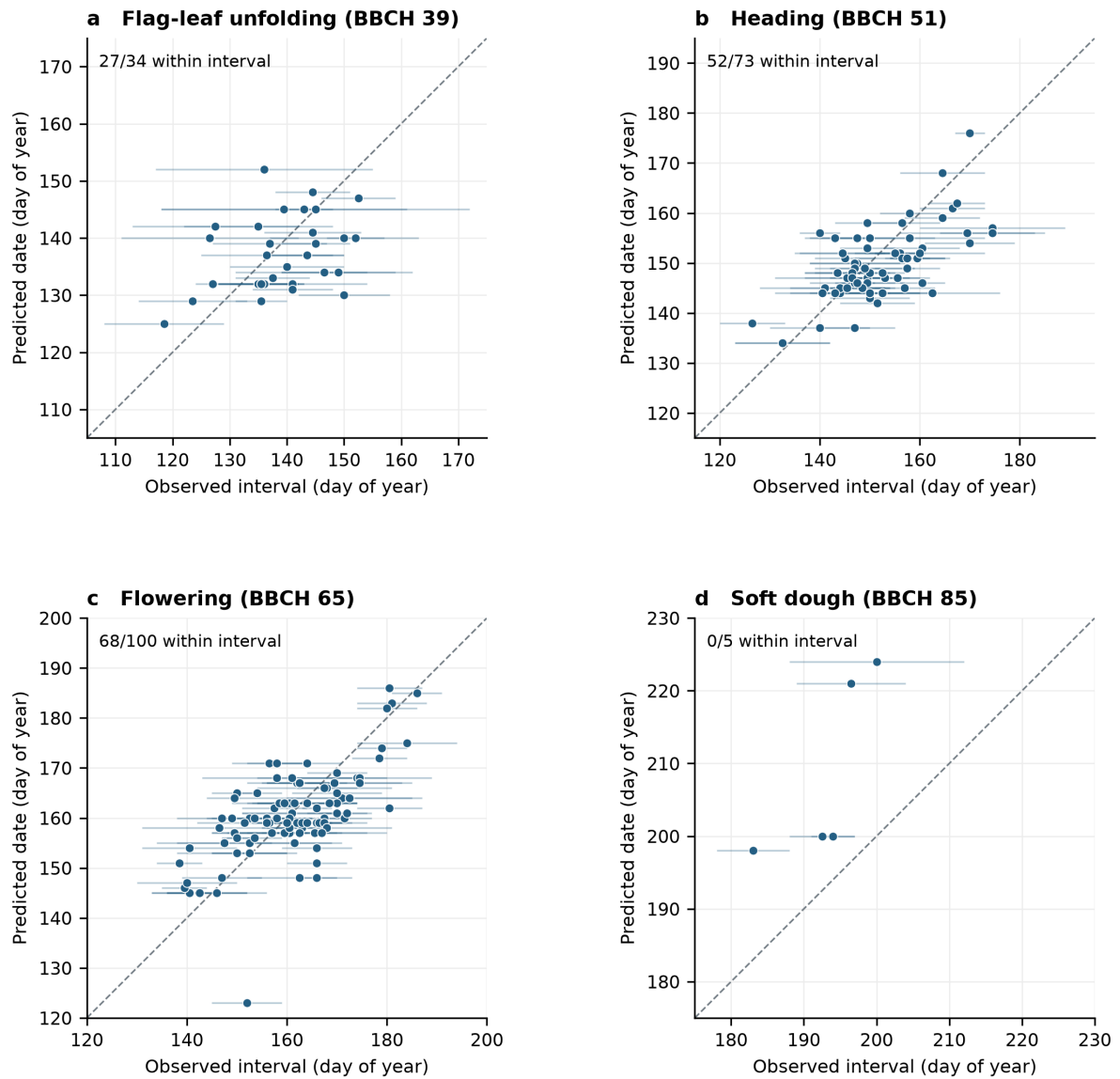


Figure S2 | Field evaluation of wheat phenological stage dates. Horizontal segments show observed two-sided stage intervals; their centers locate markers and are not treated as exact event dates. The dashed diagonal denotes equal dates, and an interval intersecting it contains the prediction. Counts give compatible intervals among all informative records. Observations from both field archives are pooled; calibration and evaluation remain separate. Intervals express observation resolution rather than statistical confidence.

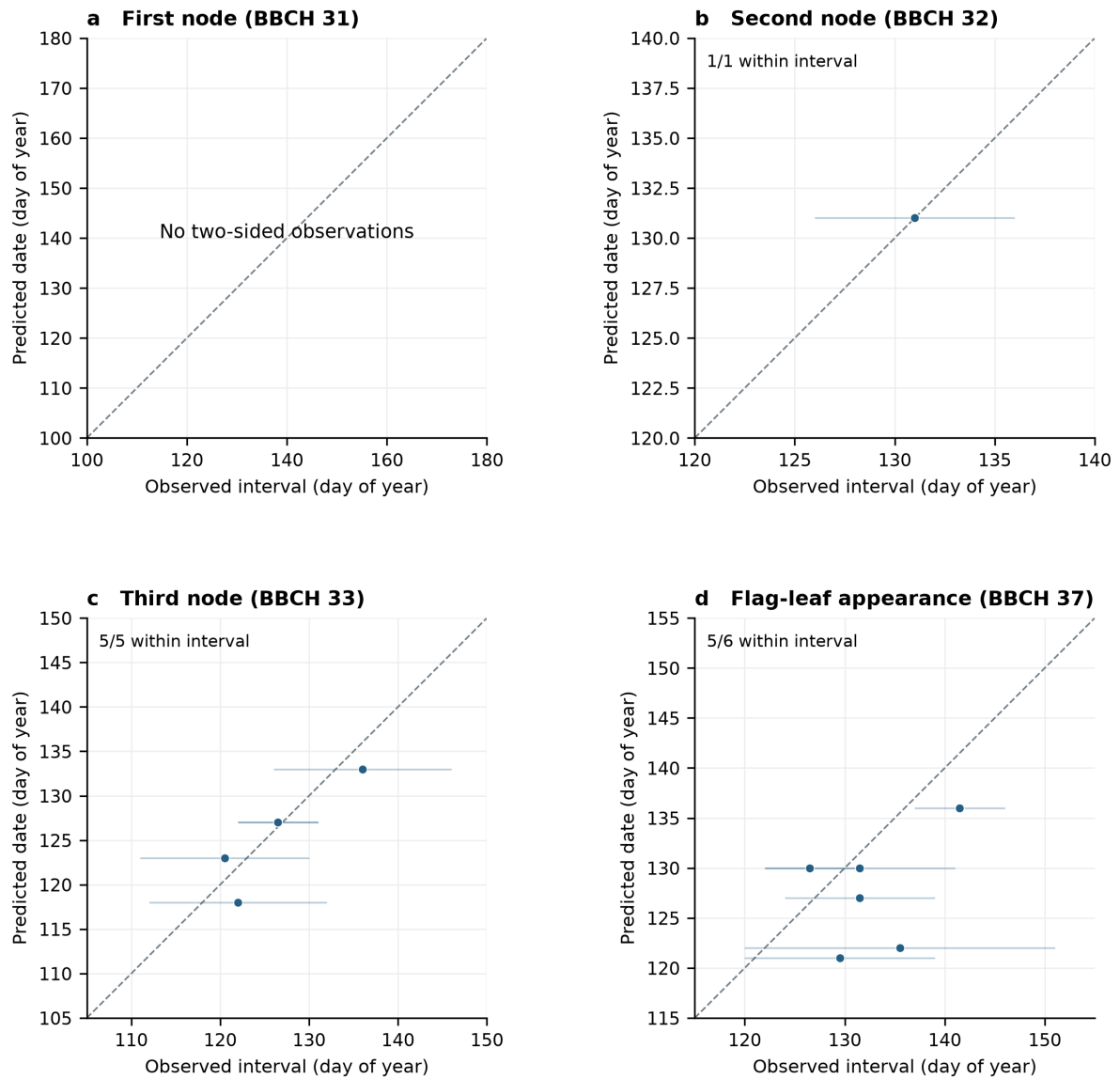


Figure S3 | Calibration of wheat phenological stage dates. Horizontal segments show observed two-sided stage intervals; their centers locate markers and are not treated as exact event dates. The dashed diagonal denotes equal dates, and an interval intersecting it contains the prediction. Counts give compatible intervals among all informative records. Observations from both field archives are pooled; calibration and evaluation remain separate. Intervals express observation resolution rather than statistical confidence.

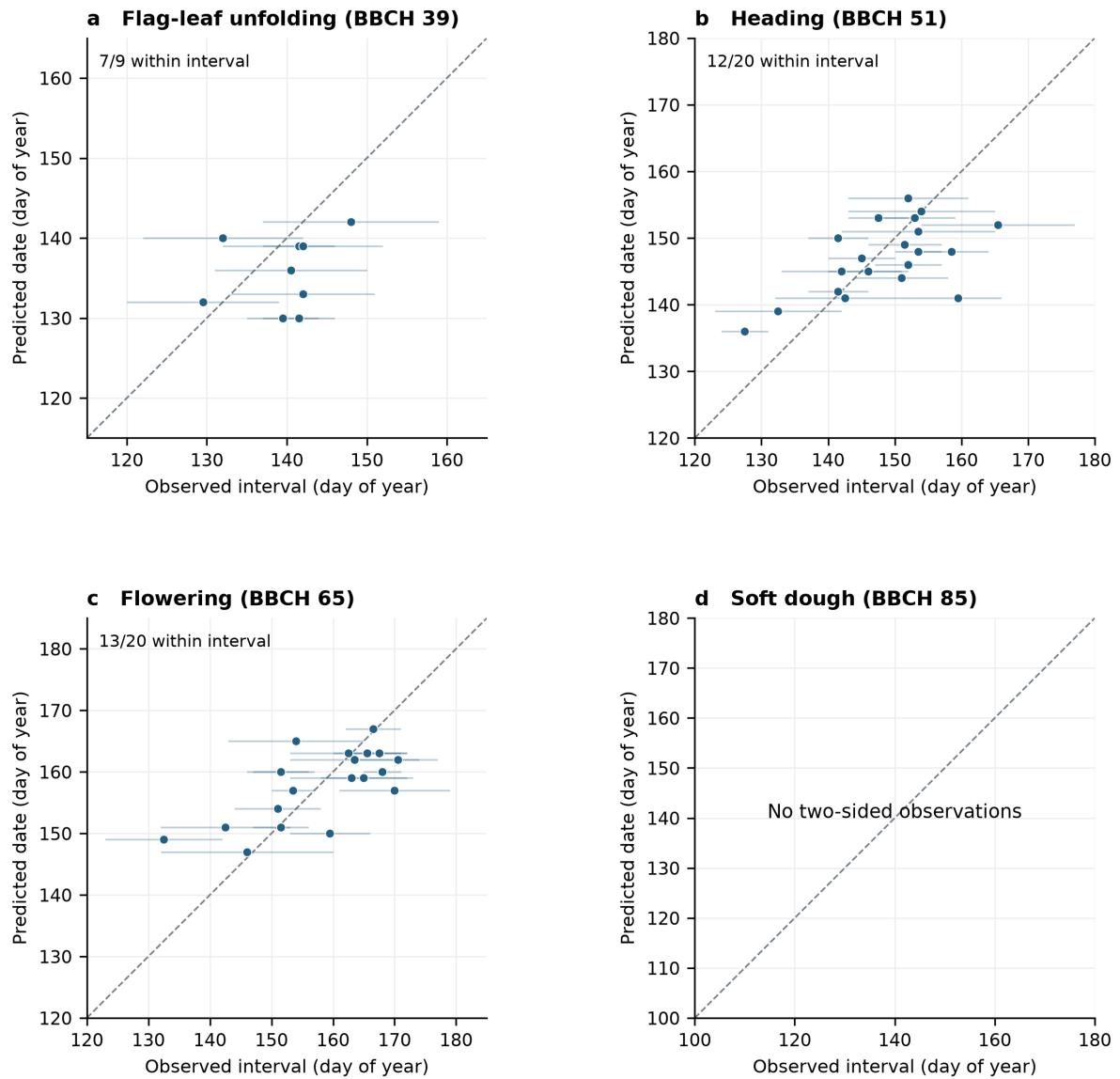


Figure S4 | Calibration of wheat phenological stage dates. Horizontal segments show observed two-sided stage intervals; their centers locate markers and are not treated as exact event dates. The dashed diagonal denotes equal dates, and an interval intersecting it contains the prediction. Counts give compatible intervals among all informative records. Observations from both field archives are pooled; calibration and evaluation remain separate. Intervals express observation resolution rather than statistical confidence.

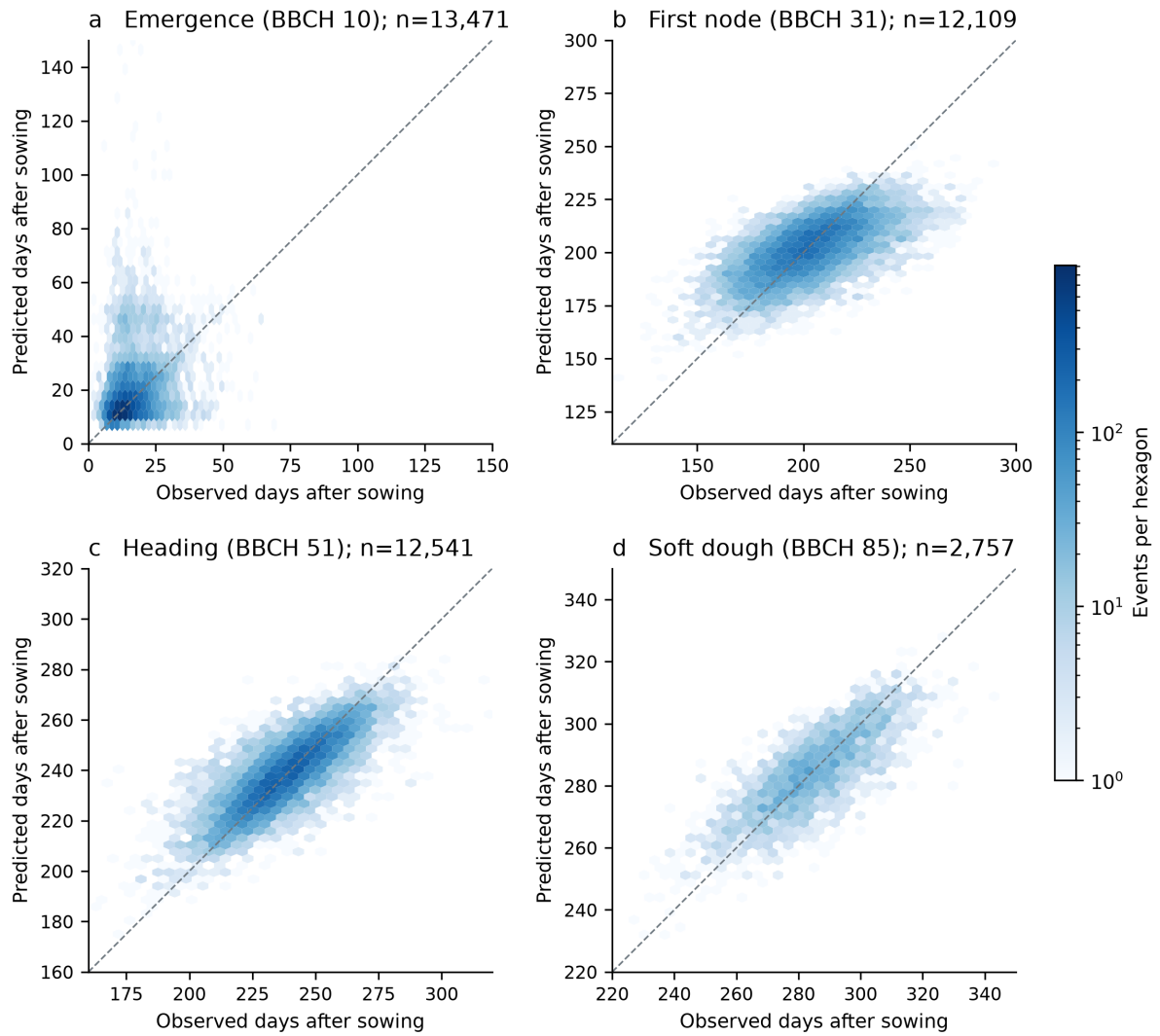


Figure S5 | Retained T-P-V predictions on the German station test cohort at BBCH 10, 31, 51 and 85. Every common matched test event is represented, including date-error tails. Both axes use days from the recorded sowing anchor to avoid calendar-year ambiguity. Hexagon colour shows count on a shared logarithmic scale; axis ranges vary by stage. Missing predictions remain in the coverage tables. The station holdout concerns realized-weather phenology, rather than disease or field-yield forecasts.

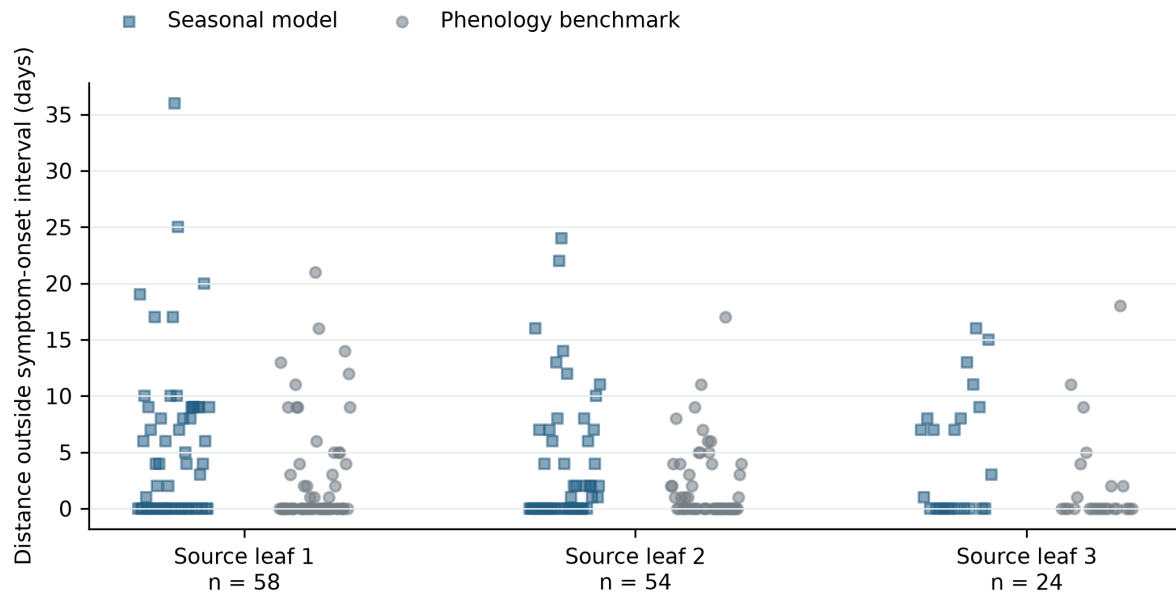


Figure S6 | Pooled symptom-onset errors for source-numbered leaves 1-3. All 136 genuinely two-sided histories remain represented. Horizontal displacement separates coincident records. Zero indicates a predicted date within the observation interval; missing positive predictions retain their predefined penalty. Source leaf numbers do not establish a common final-rank identity across archives. Scores use equal coordinate-year, field and leaf weighting.

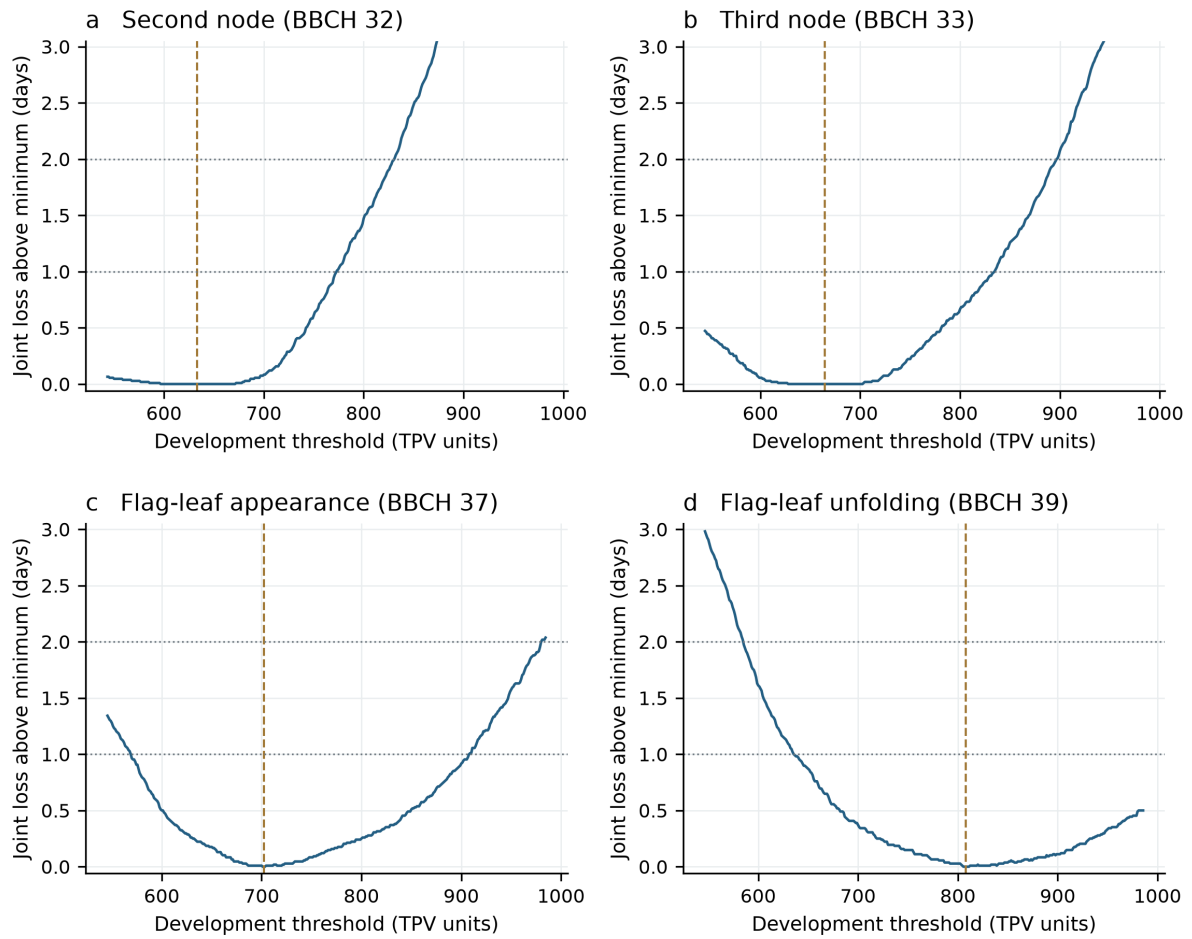


Figure S7 | Ordered stage-threshold identification from calibration-only observations. Each blue curve is the minimum joint loss conditional on that threshold, after optimizing the other ordered stage thresholds. Dashed vertical lines mark the selected values; horizontal lines show declared one- and two-day loss tolerances. The displayed range is zero to three days above the joint minimum; all 399 grid values for each event remain in Source Data. Loss-tolerance regions describe identification and are not statistical confidence intervals.

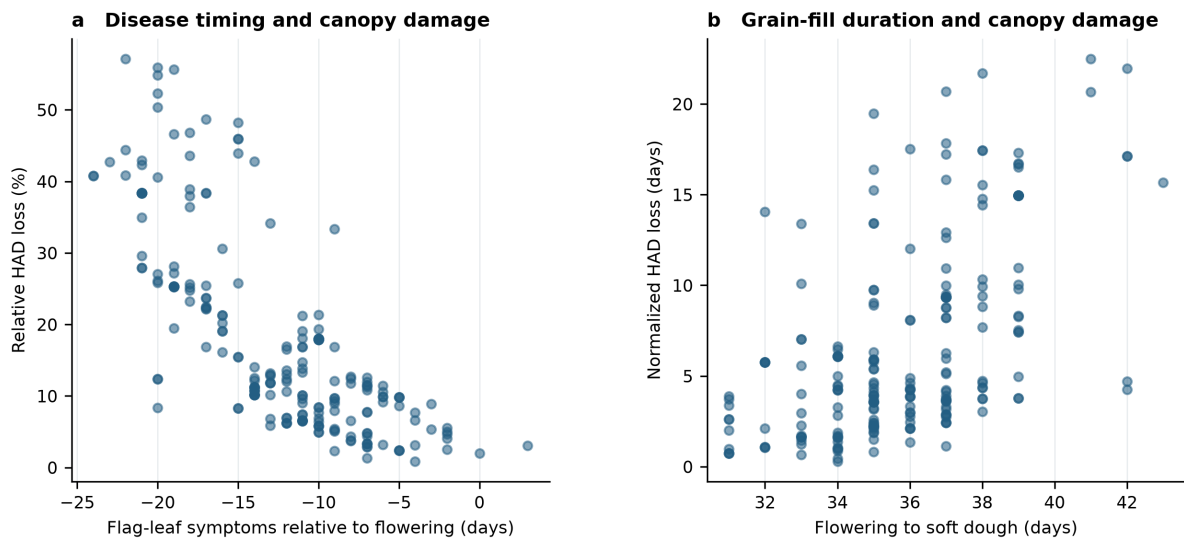


Figure S8 | Simulated disease timing and grain-fill duration in field weather replays. Each point represents one of 218 full-season simulations. (a) Flag-leaf symptoms relative to flowering and relative HAD loss. (b) Flowering-to-soft-dough duration and HAD deficit. Reference upper-three-leaf LAI is nominally one. Both axes describe simulated quantities; these relationships do not validate observed disease severity or grain-yield loss.

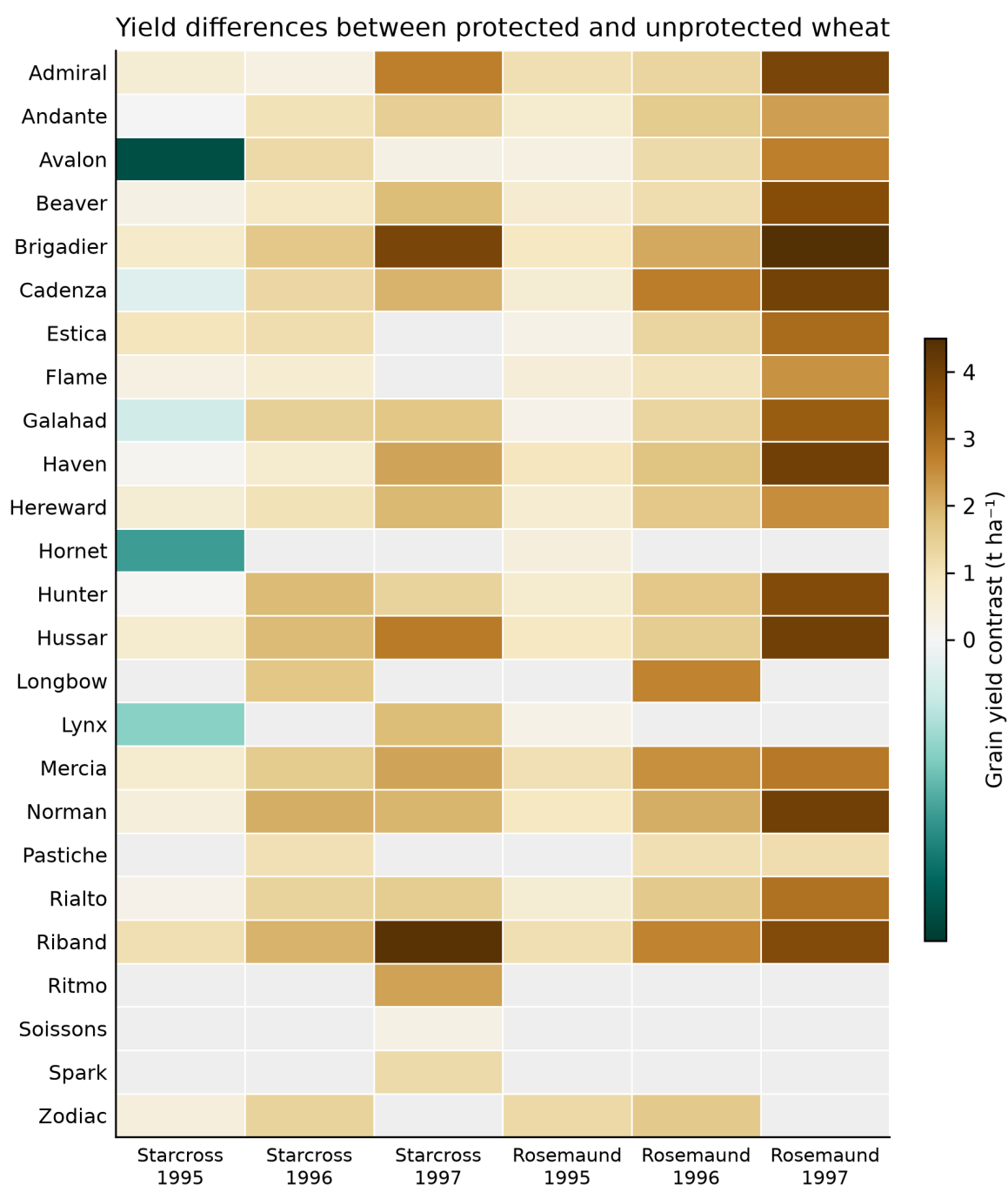


Figure S9 | All 117 reported protected-minus-unprotected grain-yield contrasts in Parker et al. (2004), Table 4. The 25 cultivars and six site-years are retained, including five negative contrasts; grey cells indicate unreported combinations. Values are published treatment contrasts from a three-replicate field design, not independent plot records. They supply empirical yield outcomes for the study whose crop-impact relationship integrates absolute green area over the top three leaves.

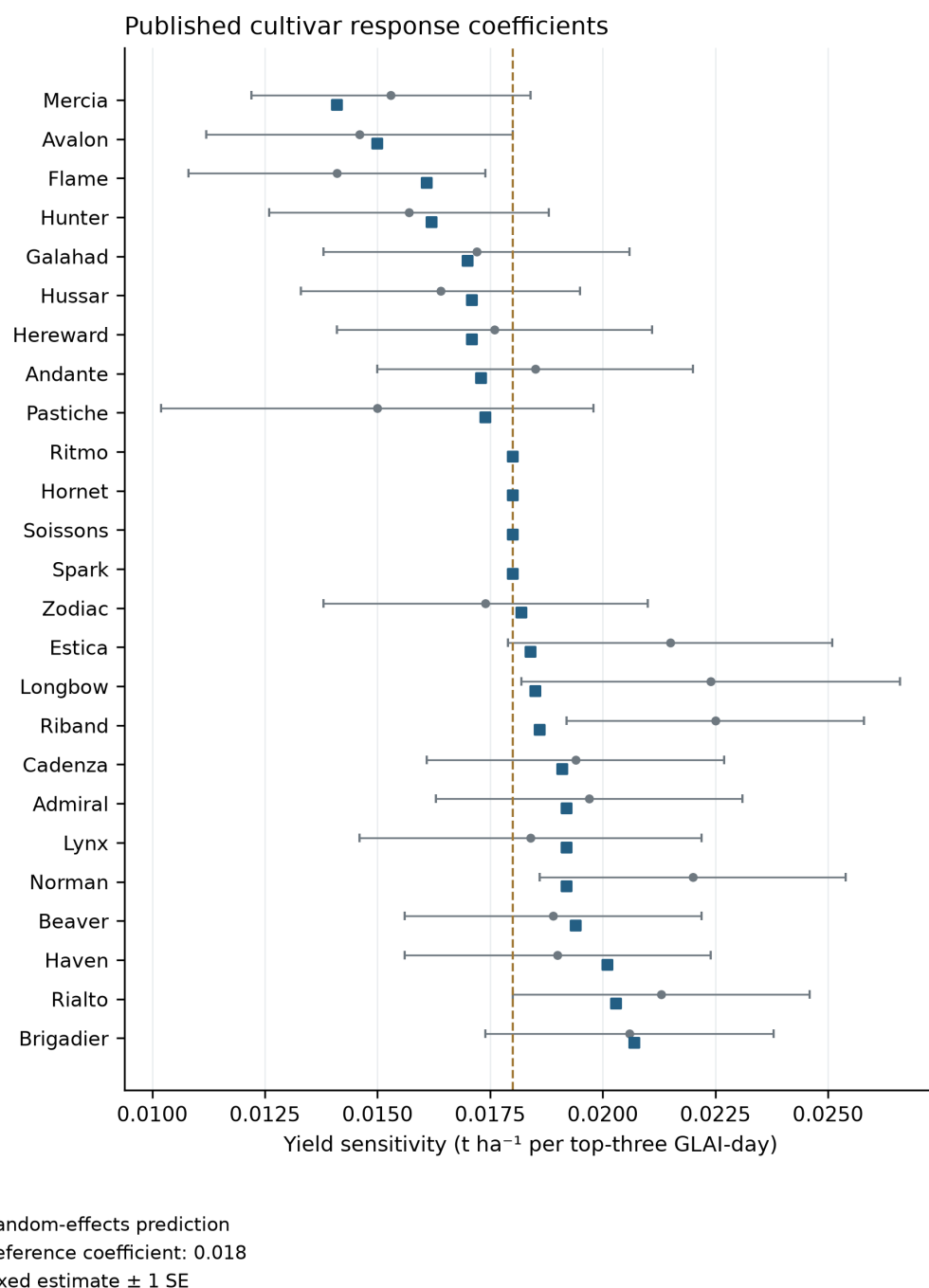


Figure S10 | All 25 cultivar response predictions from Parker et al. (2004), Table 5, with available fixed-effect estimates and their published standard errors. The dashed line identifies the retained 0.018 t ha⁻¹ per GLAI-day transfer centre. Random-effect predictions span 0.0141–0.0207; that cultivar range is not a confidence interval. Error bars represent one standard error of fixed-effect coefficients. BLUP prediction-error variances concern deviations from the population mean and are not substituted for uncertainty of complete slopes. Different cultivars share environments and are not independent field replications.

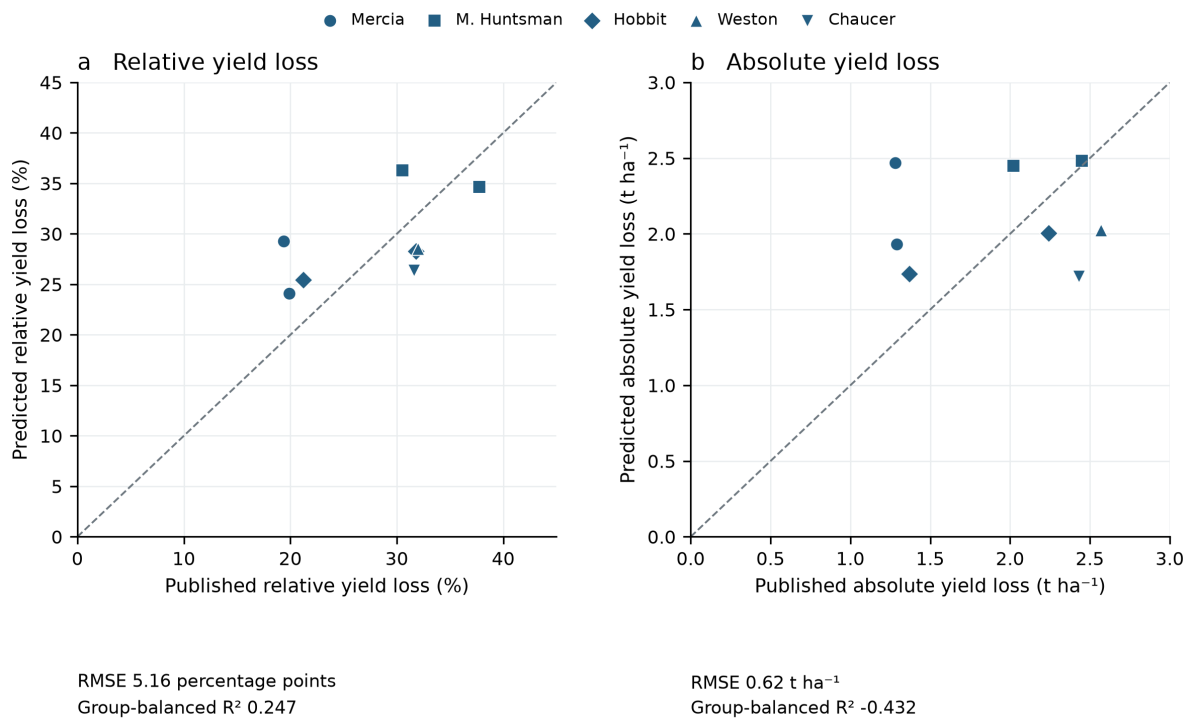


Figure S11 | Leave-one-genetic-background-out reproduction of the published top-five paired means. One nonnegative origin-constrained coefficient is fitted to the other four genetic backgrounds in each fold. (a) Relative HAD-loss to relative yield-loss transfer; (b) absolute HAD-loss to absolute grain-loss transfer. Metrics give each background equal weight, divided among its lines. This is a small aggregate-data benchmark with shared mixed-model means and unbalanced disease/site coverage. It is not raw-plot validation, does not remove environmental confounding and does not calibrate the current top-three transfer. Negative absolute-response R^2 identifies failure of a universal absolute slope.

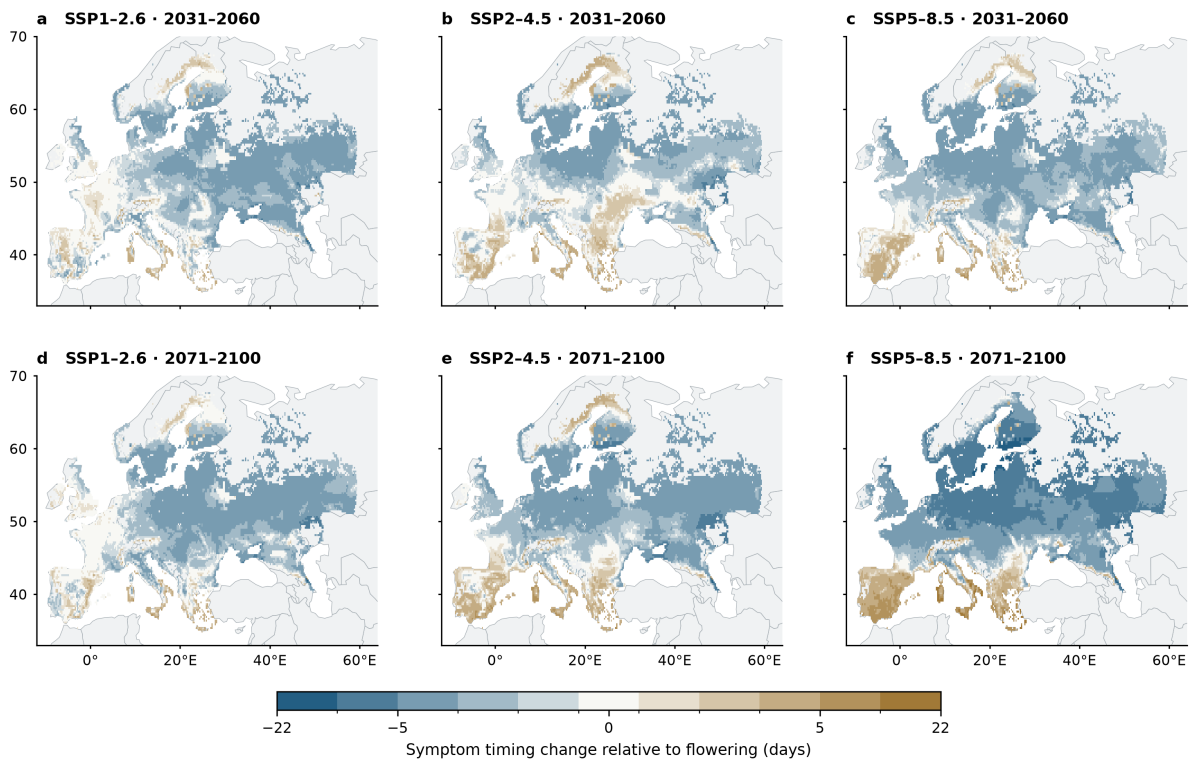


Figure S12 | Changes in flag-leaf symptom timing. Panels show three-climate-model means at every eligible wheat land-use grid for two future periods relative to 1991–2020. Each figure shares symmetric discrete color intervals retaining the full data range. Gray wheat pixels have unavailable estimates. Timing requires detected symptoms in both paired years. Estimated disease-related yield change is -0.018 times the HAD-loss change, converted to kg ha^{-1} per reference LAI. The yield product depends on a fixed canopy-yield coefficient and nominal leaf area rather than a validated grain-yield model.

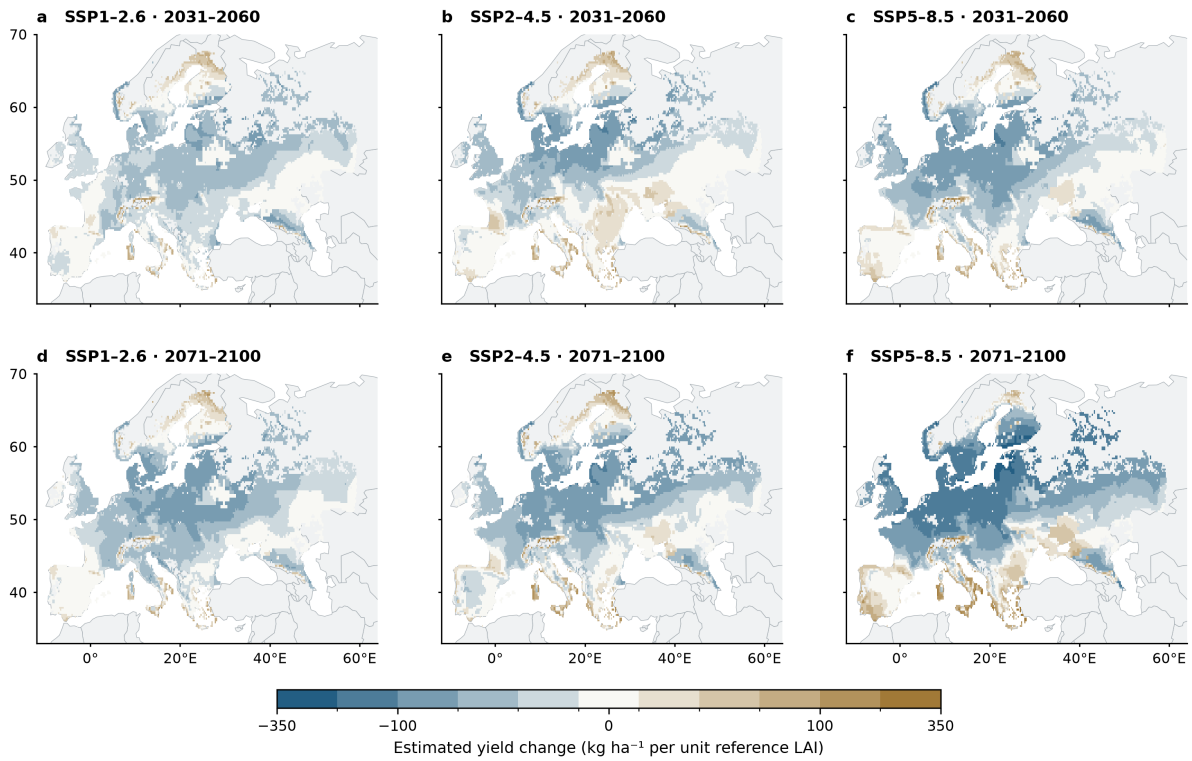


Figure S13 | Disease-related yield changes inferred from canopy damage. Panels show three-climate-model means at every eligible wheat land-use grid for two future periods relative to 1991–2020. Each figure shares symmetric discrete color intervals retaining the full data range. Gray wheat pixels have unavailable estimates. Timing requires detected symptoms in both paired years. Estimated disease-related yield change is -0.018 times the HAD-loss change, converted to kg ha⁻¹ per reference LAI. The yield product depends on a fixed canopy-yield coefficient and nominal leaf area rather than a validated grain-yield model.

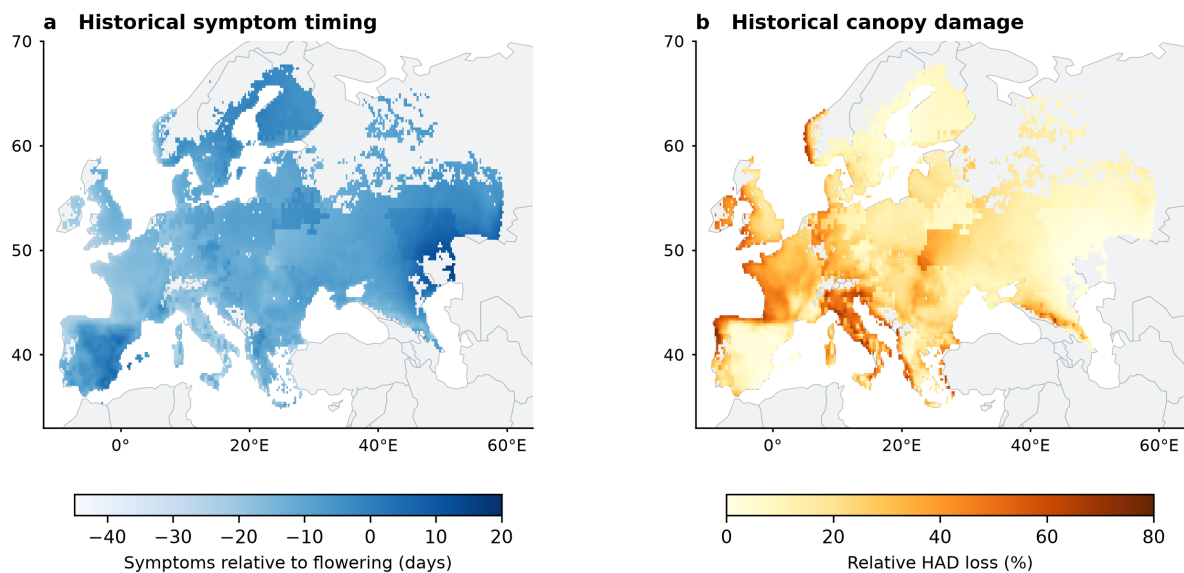


Figure S14 | Historical simulated disease timing and canopy damage across the full wheat domain. Panels show the 1991–2020 reference means on the corresponding seasons included in the late-century SSP5–8.5 comparison, averaged across three climate models. (a) Flag-leaf symptoms relative to flowering. (b) Relative HAD loss during grain filling. Gray wheat pixels have unavailable estimates. These modeled reference distributions differ from observed historical disease prevalence.

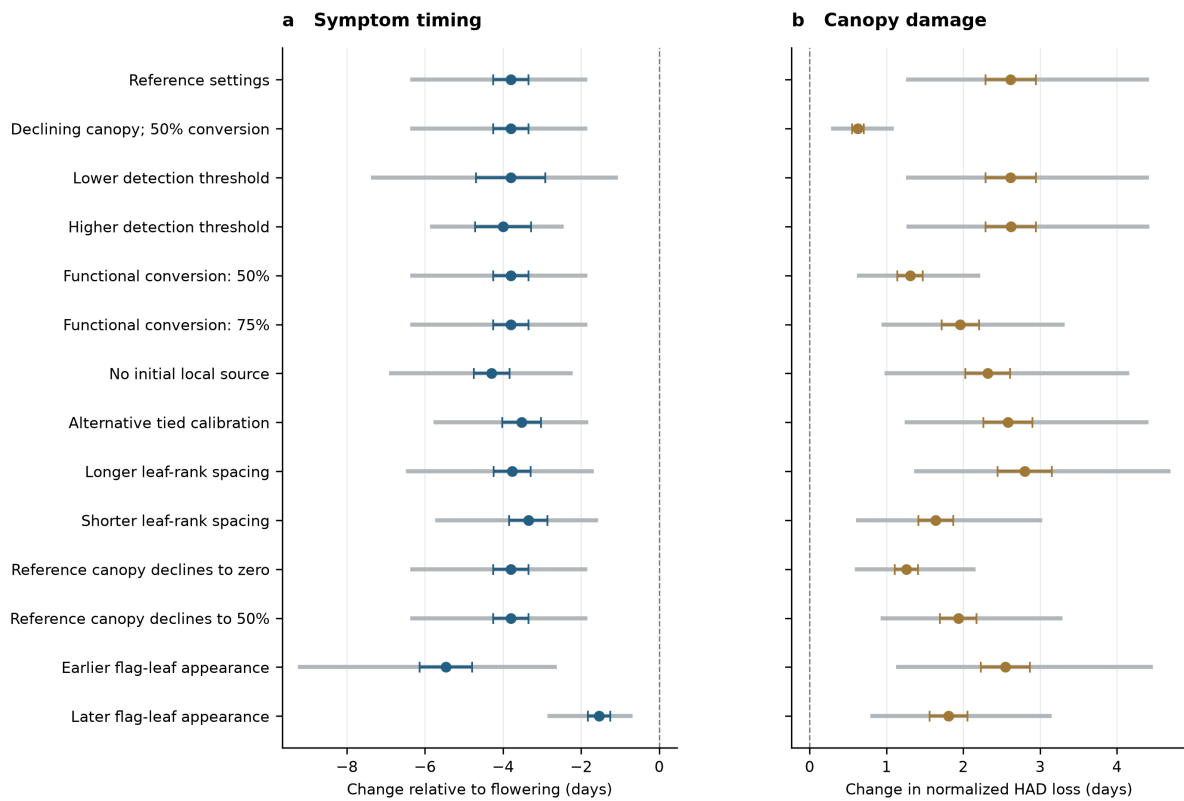


Figure S15 | Sensitivity of simulated climate responses to structural and reporting assumptions. (a) Flag-leaf symptom timing relative to flowering. (b) Normalized HAD loss during grain filling. All fourteen settings retain their common paired late-century SSP5-8.5 diagnostic population. Gray segments span three climate-model means; colored whiskers show one spatial Monte Carlo standard error for 64 area-proportional draws. Settings describe selected model assumptions rather than a probability distribution of future disease outcomes; precise settings and coverage remain in Supplementary Table S10.

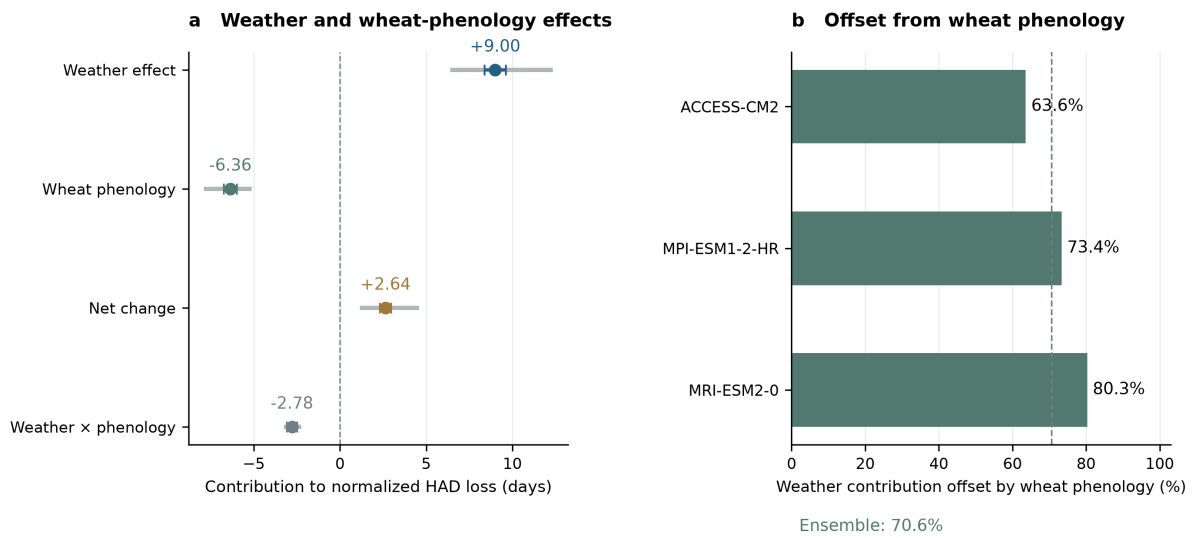
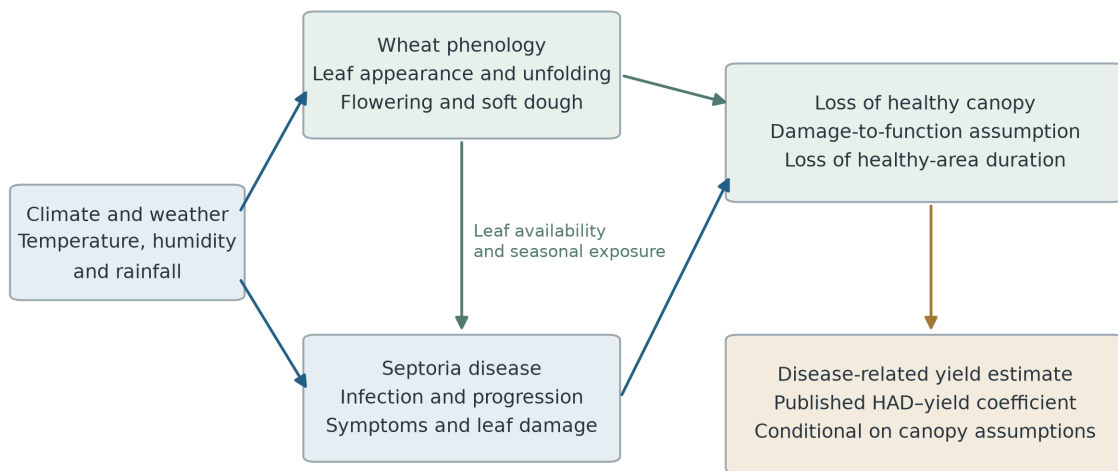


Figure S16 | Weather and wheat-phenology contributions and offsets. (a) Weather, crop-phenology and net HAD contributions with the full interaction shown separately. The interaction is already allocated equally between the two Shapley contributions and is not added again to the net. Gray segments span climate-model means; colored whiskers show one spatial Monte Carlo standard error. (b) Model-specific offsets and their ensemble-contribution ratio. The same 5,731 supported paired seasons underlie all HAD terms. The sampled analysis is separate from the full-grid estimates and does not measure intervention benefits.



Wheat phenology drives disease exposure; disease damage does not feed back into developmental rates.

Figure S17 | Relationships among climate, wheat phenology, Septoria disease and the estimated yield response. Wheat phenology comprises leaf appearance, unfolding, flowering and soft dough. Weather drives these stage dates and disease. Leaf availability and seasonal timing connect wheat phenology to infection and damage. Disease damage is mapped to an assumed loss of canopy function and integrated as HAD loss. A published coefficient provides a conditional disease-related yield estimate. The model has no disease feedback on crop-development rates and no complete crop carbon-balance calculation.

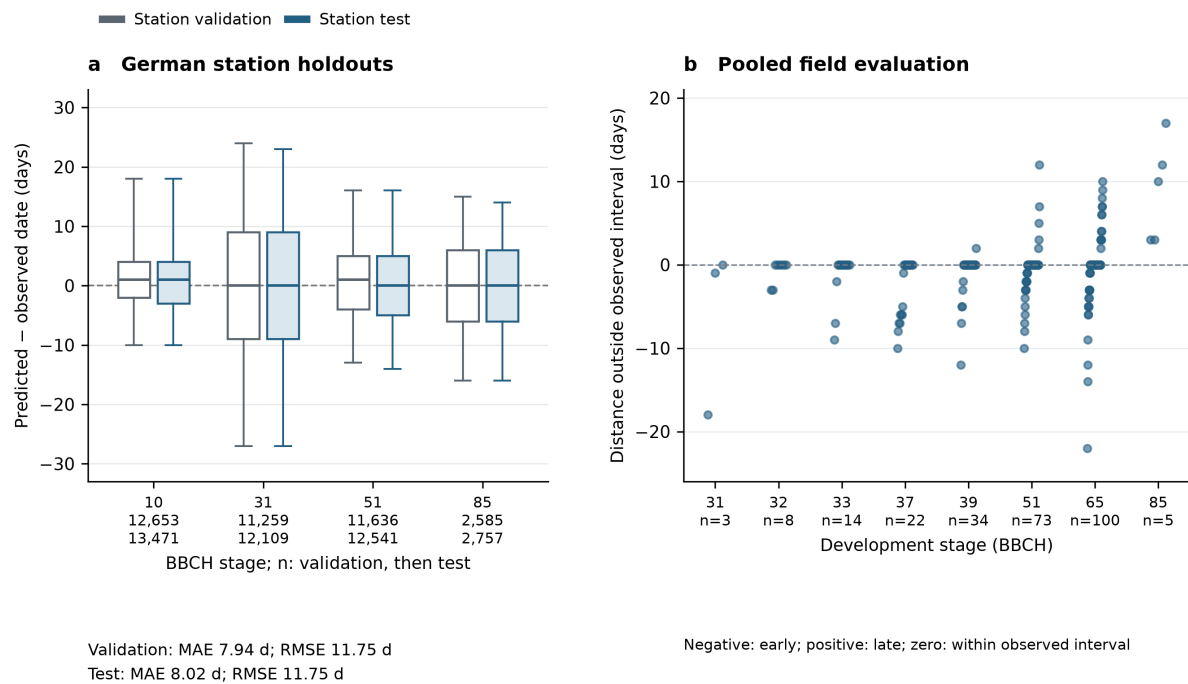


Figure S18 | Evaluation of retained crop-development stages. (a) German station-held-out date errors at BBCH 10, 31, 51 and 85. Boxes span the interquartile range; whiskers span the fifth to ninety-fifth percentiles. (b) Pooled field errors outside genuine two-sided developmental-stage intervals, with all informative records represented. The station and field endpoints have different observation resolutions. Counts beneath stages give sample sizes; observation intervals do not define statistical confidence intervals.

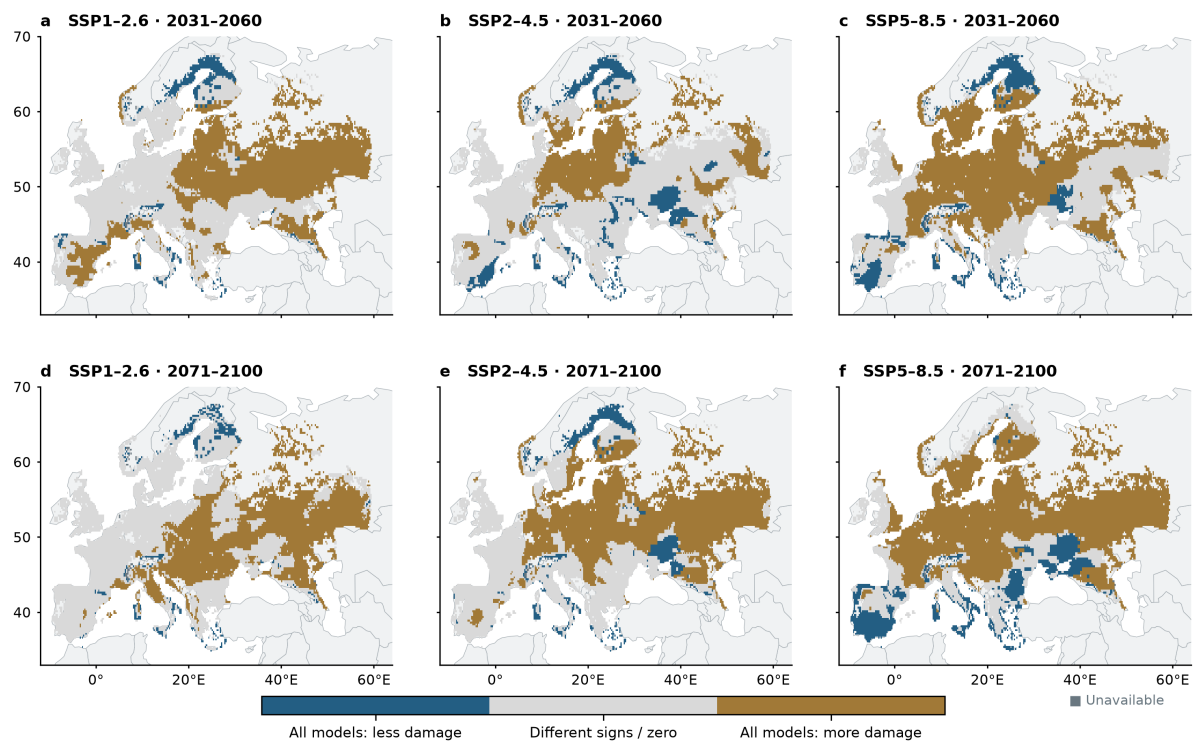


Figure S19 | Agreement in projected canopy-damage direction across climate models. Colors indicate a negative change in all three models, a positive change in all three models, or other sign combinations. Each panel shows independently simulated wheat grids for one emissions pathway and future period relative to 1991–2020. Model agreement is deterministic scenario consistency rather than a probability estimate. Unavailable wheat pixels remain distinguishable from complete mixed-sign estimates in Source Data.

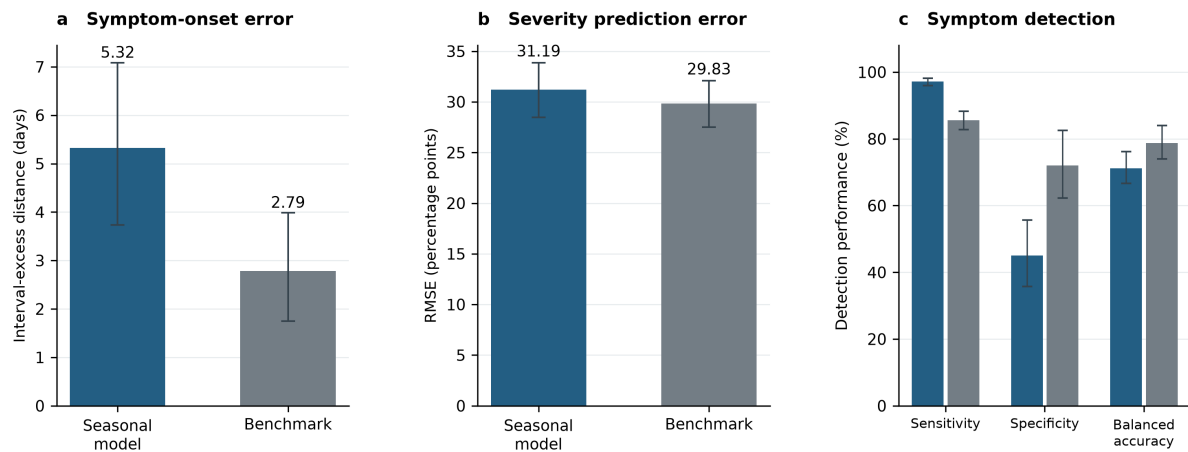


Figure S20 | Pooled evaluation of disease predictions. (a) Mean distance outside genuine two-sided symptom-onset intervals for 136 histories from 78 field seasons and 62 coordinate-years. (b) Numerical severity RMSE for 2,033 assessments from 188 field seasons and 140 coordinate-years. (c) Assessment-level symptom-detection sensitivity, specificity and balanced accuracy on the same 2,033 records. Blue bars represent the seasonal model and gray bars the corresponding calibration-selected benchmark. Whiskers are 95% percentile intervals from 20,000 paired coordinate-year bootstrap resamples. Coordinate-years, fields, source leaves and their assessments receive equal weight within each hierarchy level. Calibration records are excluded. Source scoring conventions and unresolved final leaf-rank mappings remain in the pooled evidence.

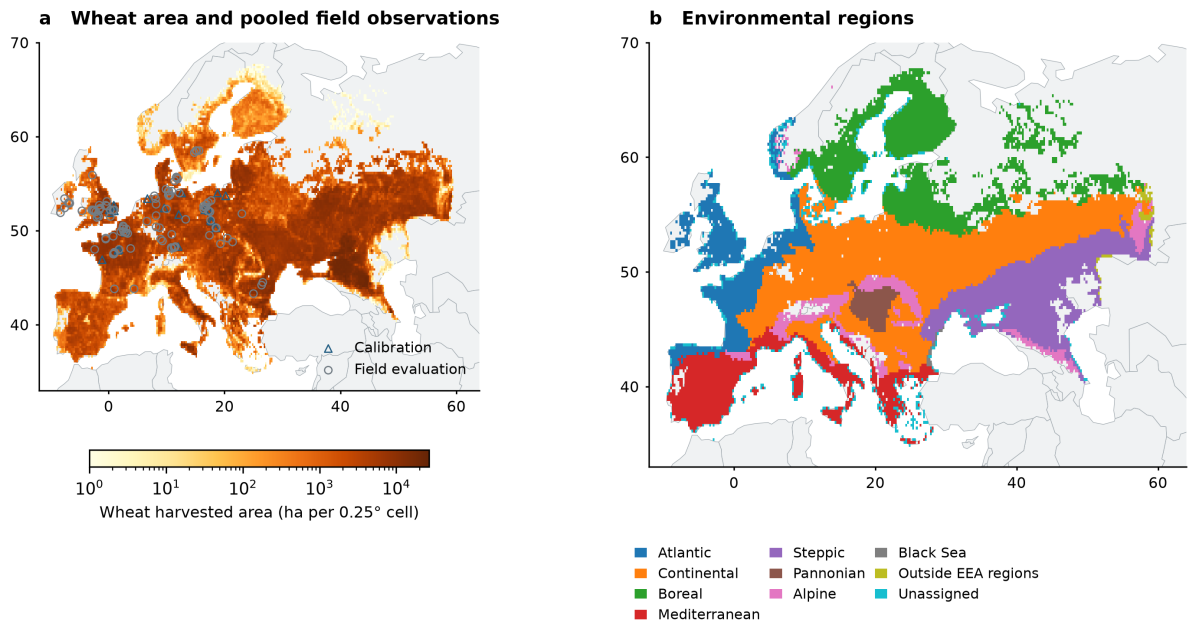
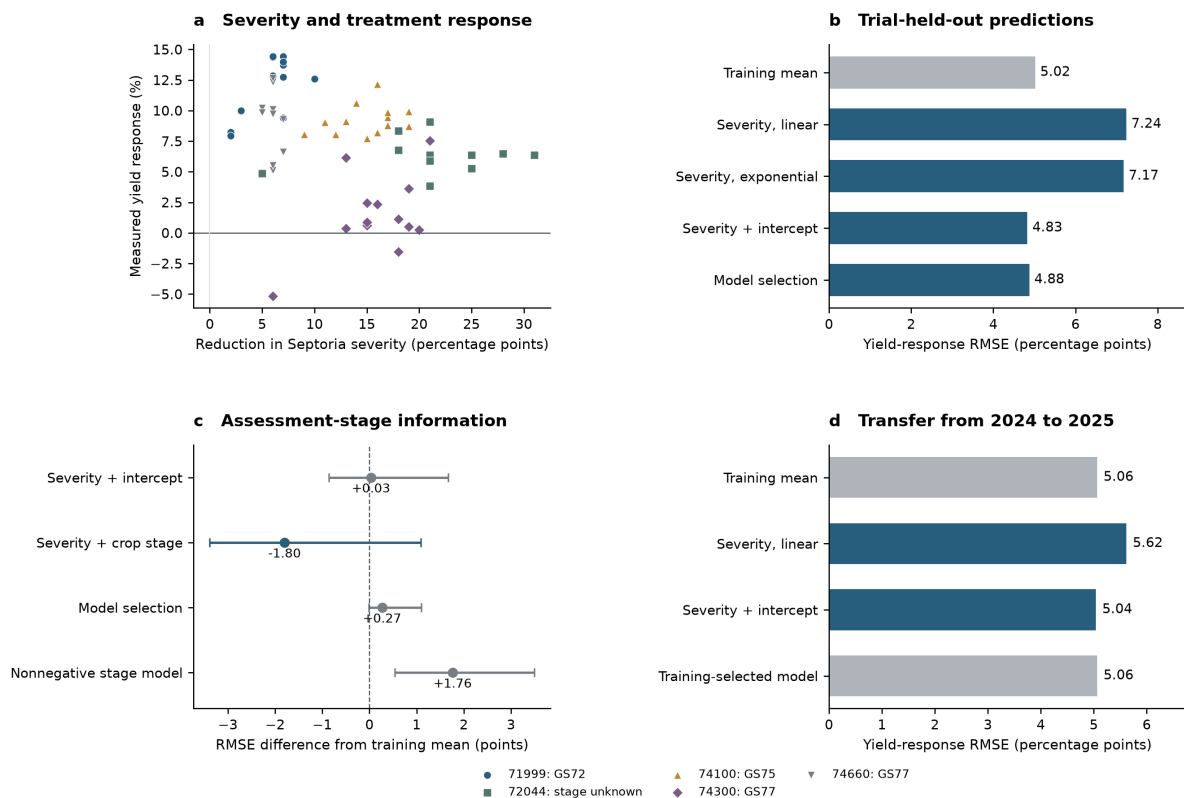
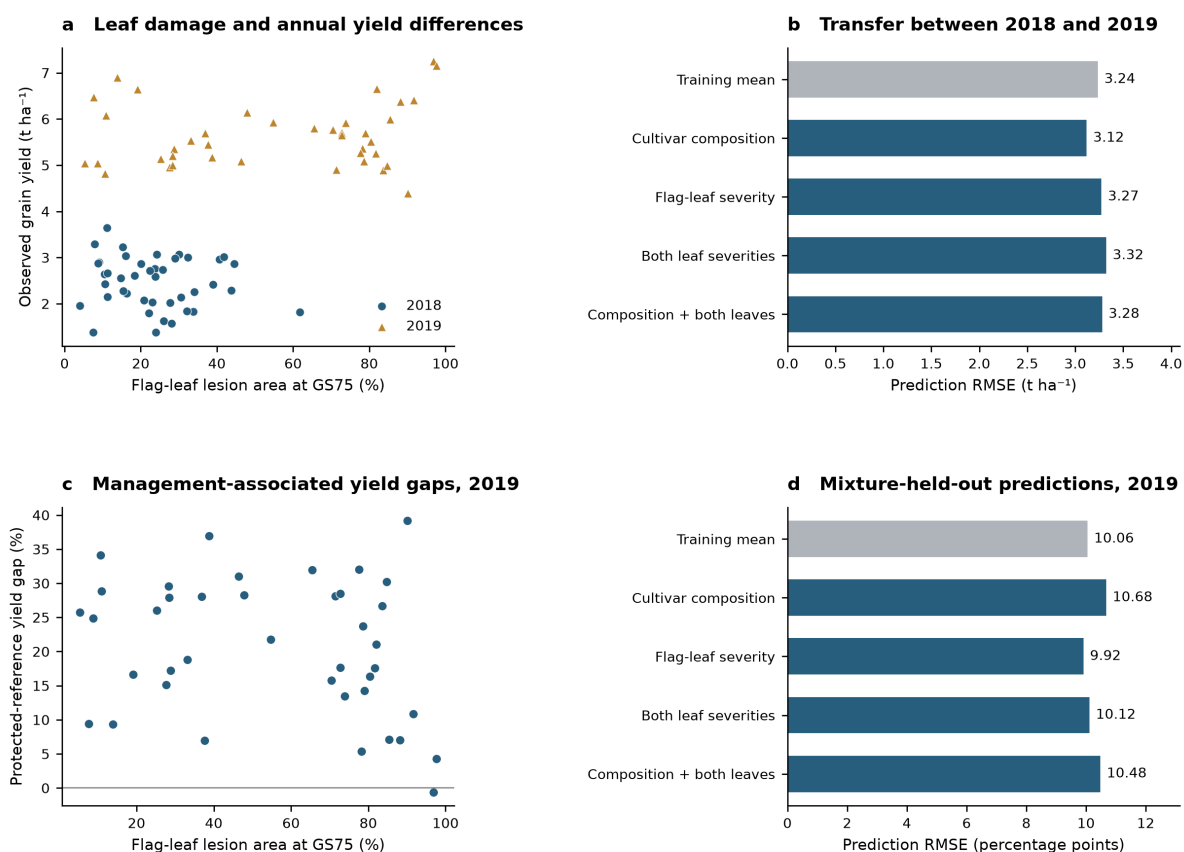


Figure S21 | Wheat-land-use domain, pooled observation locations and environmental regions. (a) SPAM2020 all-wheat harvested area with calibration and pooled evaluation locations; coincident sites retain their true reported coordinates. (b) EEA biogeographical regions assigned by wheat-cell centroid intersection. Outside-region and unassigned cells remain separate. The 14,941-cell fixed mask and imposed winter-wheat rainfed calendar define the climate-analysis scenario. These maps specify the study design and do not display projected disease outcomes.



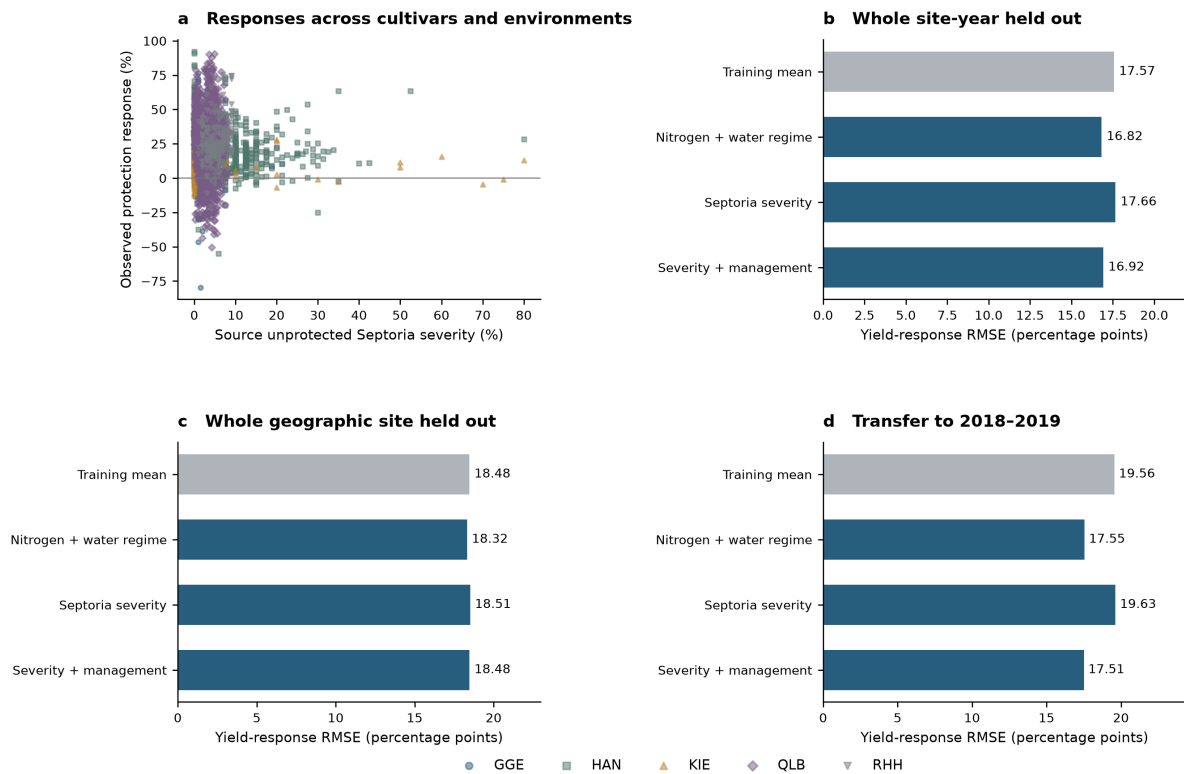
Responses are treatment contrasts; infection dates, functional leaf area and leaf-specific damage are unobserved.

Figure S22 | Nordic yield-response prediction and assessment stage. (a) Same-date severity reduction and relative treated-untreated yield response, retaining negative responses. (b) Entire-trial-held-out prediction on 62 contrasts from five trials. (c) RMSE differences from the matched training mean for the 51-contrast, four-trial population with recorded growth stage. Whiskers are 95% paired trial-bootstrap intervals conditional on fixed held-out predictions. The nonnegative stage model is a secondary sign diagnostic. (d) Forward transfer from 2024 to 2025, with 38 contrasts in three evaluation trials; the training-selected model is the training mean. The data describe protection responses against multiple diseases, rather than isolated Septoria loss or measured HAD.



Flag and lower leaves were assessed on different dates. Protected-plot severity and healthy-area duration were unobserved.

Figure S23 | Leaf-rank severity and yield-response transfer in Tunisia. (a) Flag-leaf incidence-weighted lesion area at GS75 and observed yield for 82 plot-season outcomes in 2018–2019. (b) Absolute-yield errors under two-way whole-year transfer, with years given equal weight. (c) Observed protected-reference yield gaps and unprotected flag-leaf severity in 40 paired 2019 outcomes. (d) Entire-mixture-held-out prediction in that single environment. The lower leaf was assessed at GS61 in 2018 and GS73 in 2019, on different dates from the flag leaf. Protected severity and functional leaf area are unobserved; the two leaf assessments do not establish a disease time integral. Yield moisture basis is unspecified.



Comparisons match cultivar, nitrogen and water regime. Protection targets multiple diseases; assessment dates and leaf ranks are unspecified.

Figure S24 | German multi-environment protection-response prediction. (a) Unprotected Septoria severity and observed relative protection response for 3,264 cultivar-management comparisons from 16 site-years at five locations. Negative responses remain in the data. Treatment means match cultivar, nitrogen and source-coded water regime; drought-suffixed references are excluded. (b) Whole-site-year-held-out errors, with site-years weighted equally. (c) Whole-geographic-site-held-out errors, with locations weighted equally. (d) Forward transfer from 2015-2017 to 619 comparisons in six site-years during 2018-2019. German yields share a dry-mass basis, which cancels in the relative response. Missing severity remains unknown. Protection targets multiple diseases; disease-assessment dates and final-leaf ranks are undocumented.

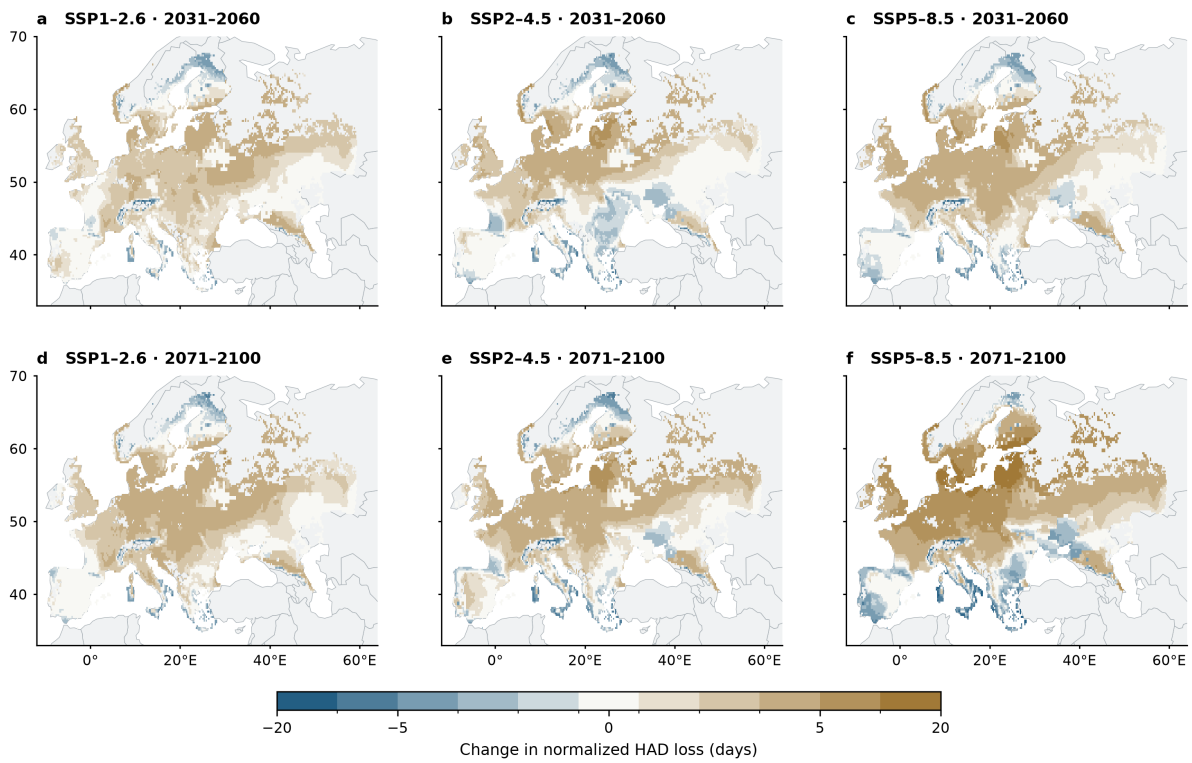


Figure S25 | Canopy-damage changes across all emissions pathways and periods. Three-climate-model mean changes in normalized healthy-area-duration loss during flowering to soft dough relative to 1991–2020, shown on the native quarter-degree wheat grid. Positive values indicate greater loss of assumed canopy function. All six panels share a symmetric colour scale. Grey wheat cells have unavailable estimates. The all-wheat production domain uses imposed winter-wheat rainfed calendars.

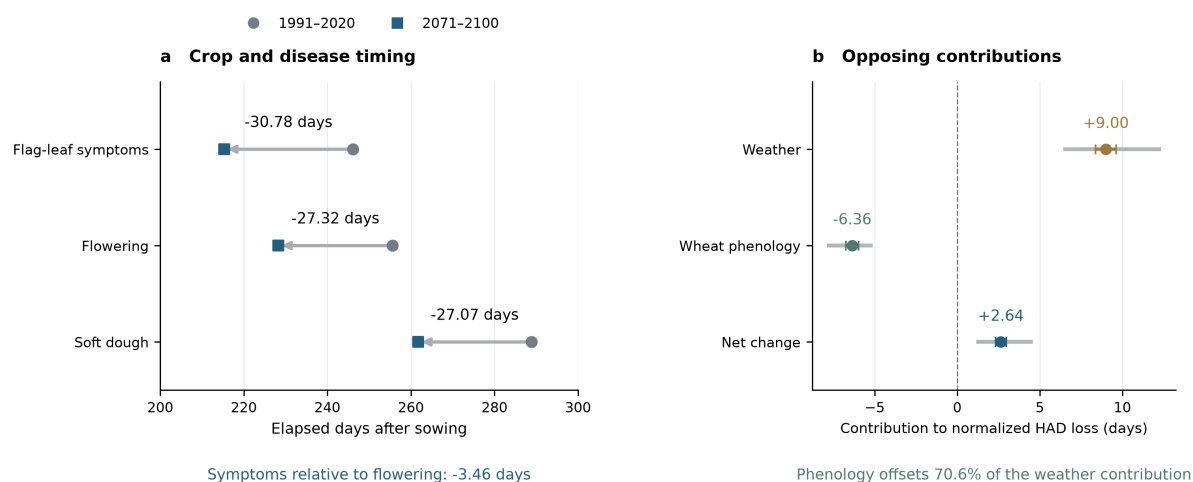


Figure S26 | Common-population calendar timing and weather-phenology contributions. (a) European mean dates under SSP5-8.5. Symptom appearance, flowering and soft dough use identical historical-future season pairs with all dates recorded in the full-grid outputs. Fixed harvested-area weighting precedes equal climate-model averaging. Arrows connect period means; the horizontal axis shows elapsed calendar days after sowing. (b) Weather, wheat-phenology and net contributions to normalized HAD loss in the separate 62-cell diagnostic sample (64 area-proportional draws; 5,731 valid season pairs). Grey segments span climate-model means; coloured whiskers show one spatial Monte Carlo standard error. Each Shapley contribution includes half the weather-phenology interaction. The 70.6% offset is the ratio of ensemble contributions. The full-grid timing population in a differs from the sampled decomposition in b. These comparisons concern climate-driven timing shifts; they do not estimate benefits of changing sowing date, cultivar or crop protection.

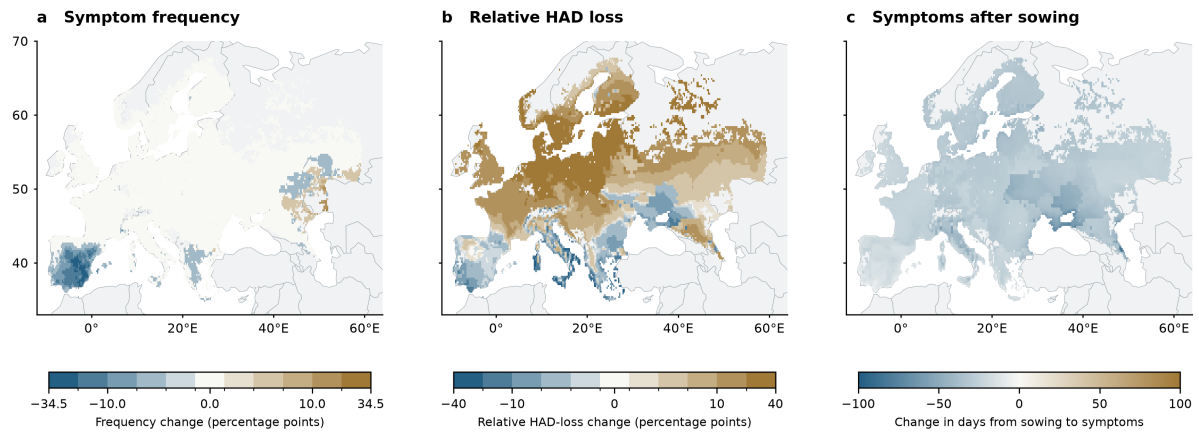


Figure S27 | Complementary symptom and canopy diagnostics under late-century SSP5-8.5. Three-model mean changes in 2071–2100 relative to 1991–2020. (a) Frequency of flag-leaf symptoms before soft dough. (b) Relative HAD loss during grain filling; this is the fraction of assumed canopy function lost, not observed lesion percentage. (c) Elapsed days from the fixed sowing date to first symptoms. Its continuous linear symmetric colour scale spans every finite change, retaining distinctions among ordinary advances and rare extremes. Date arithmetic is checked against recorded sowing and symptom dates; this consistency check does not validate biological plausibility of extreme projections. Timing requires detected symptoms in both paired seasons, whereas frequency includes complete symptom-free seasons. Grey wheat cells have unavailable estimates.